

4

Kinematics

Kinematics is the study of motion exclusive of the influences of mass, force and stress: it is the study of the geometry of motion. In this chapter we study the kinematics of continuum bodies. We focus on how to quantify strain and rate of strain in a body whose shape changes with time. In subsequent chapters we present various types of constitutive relations between stress and strain that characterize different types of materials. The contents of this chapter provide the basis for understanding the physical reasoning behind those relations, as well as the mathematical language for their precise statement.

The important ideas in this chapter are: (i) the notion of a deformation and a deformation map; (ii) the distinction between material and spatial coordinates; (iii) the notion of strain and its quantification; (iv) the idea of a motion and the material time derivative; (v) the notion of rate of strain and its quantification; (vi) the change of variable formulae for integration over material and spatial coordinates; (vii) special types of deformations and motions which preserve the shape and/or volume of a body.

4.1 Configurations and Deformations

At any instant of time a material body occupies an open subset B of Euclidean space $\mathbf{E}^3$ as described in Section 3.1. The identification of material particles with points of B defines the **configuration** of the body at that instant. We usually refer to the subset B as the configuration, and consider the identification of particles with points to be implicitly understood.

By a **deformation** we mean a change of configuration between some

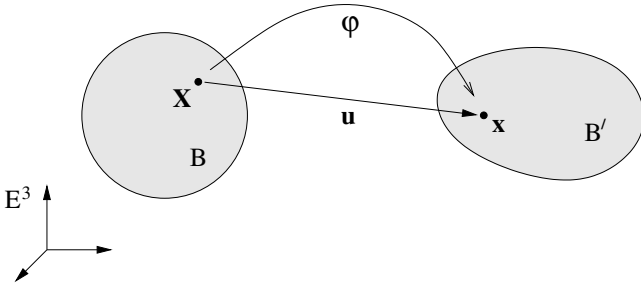


Fig. 4.1 Illustration of a deformation: B denotes the reference configuration and B' denotes the deformed configuration. The black dot in B and B' represents one and the same material particle.

initial (undeformed) configuration B and a subsequent (deformed) configuration B' (see Figure 4.1). By convention, we call B the **reference configuration** and B' the **deformed configuration**. We denote points in B by $\mathbf{X} = (X_1, X_2, X_3)^T$ and points in B' by $\mathbf{x} = (x_1, x_2, x_3)^T$. Because B and B' are two configurations of a single material body, each particle in the body has two sets of coordinates: a set of **material coordinates** X_i for its location in B , and a set of **spatial coordinates** x_i for its location in B' .

We usually think of a deformation as changing the shape of a body. However, this need not be the case. For example, the rotation of a solid ball about its center is a deformation that does not change the shape of the ball: it occupies the same region of space, but its material particles have changed location within that region. Thus a deformation can take place even when $B' = B$. It will sometimes be useful to assume that a material body occupies all of space. In this case we always have $B' = B = E^3$.

4.2 The Deformation Map

The deformation of a body from a configuration B onto another configuration B' is described by a function $\varphi : B \rightarrow B'$, which maps each point $\mathbf{X} \in B$ to a point $\mathbf{x} = \varphi(\mathbf{X}) \in B'$ (see Figure 4.1). We call φ the **deformation map** relative to the reference configuration B . The displacement of a material particle from its initial location $\mathbf{X}$ to its final location $\mathbf{x}$ is given by

$$\mathbf{u}(\mathbf{X}) = \varphi(\mathbf{X}) - \mathbf{X}. \quad (4.1)$$

We call $\mathbf{u} : B \rightarrow \mathcal{V}$ the **displacement field** associated with φ (see Exercises 1 and 2).

Certain conditions must be imposed on the map φ in order for it to represent the deformation of a material body. In particular, we assume: (i) $\varphi : B \rightarrow B'$ is one-to-one, and (ii) $\det \nabla \varphi(\mathbf{X}) > 0$ for all $\mathbf{X} \in B$. Deformations satisfying these assumptions are called **admissible**. Assumption (i) implies that two or more distinct points from B cannot simultaneously occupy the same position in B' . Assumption (ii) implies that deformations should preserve the orientation of a body; in particular, a body cannot be continuously deformed onto its mirror image (see Exercises 3 and 4). Since it is both one-to-one and onto, an admissible deformation φ is a bijection between B and B' .

Throughout the remainder of our developments we assume that all deformations are admissible. Furthermore, we assume that all deformations are smooth in the sense that partial derivatives of all orders exist and are continuous. This is always stronger than what we need, but allows for a clean presentation.

4.3 Measures of Strain

Consider an open ball Ω of radius $\alpha > 0$ centered at a point $\mathbf{X}_0$ in B as shown in Figure 4.2. Under a deformation φ , the point $\mathbf{X}_0 \in B$ is mapped to a point $\mathbf{x}_0 \in B'$, and the ball $\Omega \subset B$ is mapped onto a region $\Omega' \subset B'$. In particular, we have $\mathbf{x}_0 = \varphi(\mathbf{X}_0)$ and

$$\Omega' = \{\mathbf{x} \in B' \mid \mathbf{x} = \varphi(\mathbf{X}), \mathbf{X} \in \Omega\},$$

which, for brevity, we abbreviate as $\Omega' = \varphi(\Omega)$. Any relative difference in shape between Ω' and Ω in the limit $\alpha \rightarrow 0$ is called **strain** at $\mathbf{X}_0$. Intuitively, strain refers to the local stretching of a body caused by a deformation φ . The concept of strain plays a central role in the study of solids, and in later chapters we will see various generalizations of Hooke's Law for elastic solids which relate stress to strain. In the following sections we introduce various measures of strain and study each in some detail.

4.3.1 The Deformation Gradient $\mathbf{F}$

A natural way to quantify strain is through the **deformation gradient**, which is a second-order tensor field $\mathbf{F} : B \rightarrow \mathcal{V}^2$ defined by

$$\mathbf{F}(\mathbf{X}) = \nabla \varphi(\mathbf{X}).$$

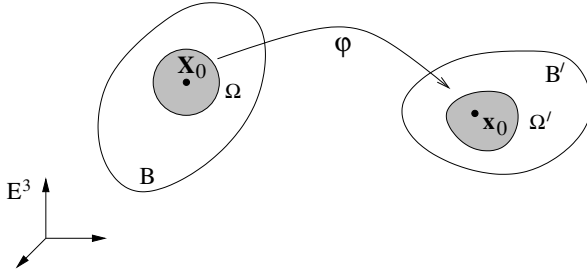


Fig. 4.2 A small region Ω in B is mapped onto a small region Ω' in B' by a deformation φ . The difference in shape between Ω' and Ω leads to the concept of strain.

The field $\mathbf{F}$ provides information on the local behavior of a deformation φ . In particular, given $\mathbf{X}_0 \in B$ we can use a Taylor expansion to write

$$\varphi(\mathbf{X}) = \varphi(\mathbf{X}_0) + \mathbf{F}(\mathbf{X}_0)(\mathbf{X} - \mathbf{X}_0) + \mathcal{O}(|\mathbf{X} - \mathbf{X}_0|^2),$$

or equivalently

$$\varphi(\mathbf{X}) = \mathbf{c} + \mathbf{F}(\mathbf{X}_0)\mathbf{X} + \mathcal{O}(|\mathbf{X} - \mathbf{X}_0|^2), \quad (4.2)$$

where $\mathbf{c} = \varphi(\mathbf{X}_0) - \mathbf{F}(\mathbf{X}_0)\mathbf{X}_0$. Thus $\mathbf{F}(\mathbf{X}_0)$ characterizes the local behavior of $\varphi(\mathbf{X})$ for all points $\mathbf{X}$ in a neighborhood of $\mathbf{X}_0$.

4.3.2 Interpretation of $\mathbf{F}$, Homogeneous Deformations

Here we describe the way in which $\mathbf{F}$ provides a measure of strain. For simplicity, we initially assume that the deformation φ is **homogeneous**, which means that the deformation gradient field $\mathbf{F}$ is constant. In this case, the higher-order terms in (4.2) vanish and φ can be written as

$$\varphi(\mathbf{X}) = \mathbf{c} + \mathbf{F}\mathbf{X}. \quad (4.3)$$

The case when the deformation φ is not homogeneous will be discussed at the end of this section (see Exercises 5 and 6).

By studying the homogeneous case we will reveal how $\mathbf{F}$ quantifies the amount of stretch and rotation experienced by a body when it is deformed from a configuration B onto a configuration B' . Due to the linear nature of (4.3), we notice that any line segment in B is mapped to a corresponding line segment in B' . This fact is exploited in the figures below (see Exercises 7 and 8).

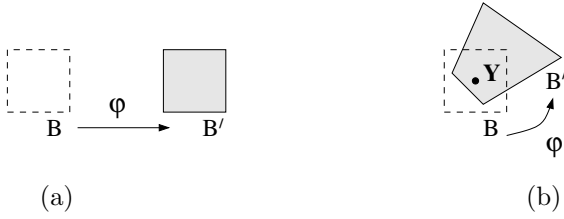


Fig. 4.3 Illustration of two homogeneous deformations. (a) translation. (b) deformation with a fixed point $\mathbf{Y}$.

4.3.2.1 Translations and Fixed Points

A homogeneous deformation φ is called a **translation** if

$$\varphi(\mathbf{X}) = \mathbf{X} + \mathbf{c}$$

for some constant vector $\mathbf{c}$. In such a deformation, each point in the body is translated or displaced along the vector $\mathbf{c}$, and there is no change in the shape and orientation of the body, as illustrated in Figure 4.3(a).

A homogeneous deformation φ is said to have a **fixed point** at $\mathbf{Y}$ if

$$\varphi(\mathbf{X}) = \mathbf{Y} + \mathbf{F}(\mathbf{X} - \mathbf{Y}). \quad (4.4)$$

The point $\mathbf{Y}$ is fixed in the sense that $\varphi(\mathbf{Y}) = \mathbf{Y}$. Any other point $\mathbf{X}$ is displaced by an amount determined by $\mathbf{F}$ and the relative position of $\mathbf{X}$ with respect to $\mathbf{Y}$. In such a deformation, the body may change both shape and orientation, as illustrated in Figure 4.3(b). We typically assume that $\mathbf{Y}$ is a point in B . However, this need not be the case.

The following result shows that an arbitrary homogeneous deformation can always be expressed as the composition of a translation and a deformation with a given fixed point. Below and throughout we use the symbol $\circ$ to denote the composition of maps.

Result 4.1 Translation-Fixed Point Decomposition. *Let φ be an arbitrary homogeneous deformation with constant deformation gradient $\mathbf{F}$. Then, given any point $\mathbf{Y}$, we can decompose φ as*

$$\varphi = \mathbf{d}_1 \circ \mathbf{g} = \mathbf{g} \circ \mathbf{d}_2,$$

where $\mathbf{d}_1$ and $\mathbf{d}_2$ are translations and $\mathbf{g}$ is a homogeneous deformation with a fixed point at $\mathbf{Y}$, in particular

$$\mathbf{g}(\mathbf{X}) = \mathbf{Y} + \mathbf{F}(\mathbf{X} - \mathbf{Y}).$$

□



Fig. 4.4 Illustration of two homogeneous deformations with fixed points. (a) rotation. (b) stretch.

Proof See Exercise 9. □

Since translations of a body are easy to understand, we concentrate on homogeneous deformations which exclude translations, that is, deformations of the form (4.4) with a fixed point $\mathbf{Y}$. Our goal is to show that (4.4) may be completely understood in simple geometric terms provided the polar decomposition of $\mathbf{F}$ is known (see Result 1.12).

4.3.2.2 Rotations and Stretches

A homogeneous deformation φ is called a **rotation** about a fixed point $\mathbf{Y}$ if

$$\varphi(\mathbf{X}) = \mathbf{Y} + \mathbf{Q}(\mathbf{X} - \mathbf{Y})$$

for some rotation tensor $\mathbf{Q}$. In such a deformation, the orientation of the body is changed holding $\mathbf{Y}$ fixed, but there is no change in the shape of the body, as illustrated in Figure 4.4(a).

A homogeneous deformation φ is called a **stretch** from $\mathbf{Y}$ if

$$\varphi(\mathbf{X}) = \mathbf{Y} + \mathbf{S}(\mathbf{X} - \mathbf{Y})$$

for some symmetric, positive-definite tensor $\mathbf{S}$. In such a deformation, the body is extended by different amounts in different directions holding $\mathbf{Y}$ fixed, but there is no net change in the overall orientation of the body, as illustrated in Figure 4.4(b).

The following result shows that an arbitrary homogeneous deformation with a given fixed point can always be decomposed into a rotation and a stretch about the same fixed point.

Result 4.2 Stretch-Rotation Decomposition. *Let φ be an arbitrary homogeneous deformation with a fixed point $\mathbf{Y}$ and deformation gradient $\mathbf{F}$ as in (4.4), and let $\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$ be the right and left*



Fig. 4.5 Illustration of homogeneous extensions along a direction $\mathbf{e}$. (a) $\lambda > 1$. (b) $0 < \lambda < 1$.

polar decompositions given in Result 1.12. Then φ can be decomposed as

$$\varphi = \mathbf{r} \circ \mathbf{s}_1 = \mathbf{s}_2 \circ \mathbf{r},$$

where $\mathbf{r} = \mathbf{Y} + \mathbf{R}(\mathbf{X} - \mathbf{Y})$ is a rotation about $\mathbf{Y}$, and $\mathbf{s}_1 = \mathbf{Y} + \mathbf{U}(\mathbf{X} - \mathbf{Y})$ and $\mathbf{s}_2 = \mathbf{Y} + \mathbf{V}(\mathbf{X} - \mathbf{Y})$ are stretches from $\mathbf{Y}$. □

Proof See Exercise 10. □

4.3.2.3 Extensions

A homogeneous deformation φ is called an **extension** at $\mathbf{Y}$ in the direction of a unit vector $\mathbf{e}$ if

$$\varphi(\mathbf{X}) = \mathbf{Y} + \mathbf{F}(\mathbf{X} - \mathbf{Y}) \quad \text{and} \quad \mathbf{F} = \mathbf{I} + (\lambda - 1)\mathbf{e} \otimes \mathbf{e}$$

for some number $\lambda > 0$. This terminology is based on the observation that $\mathbf{F}$ extends any vector parallel to $\mathbf{e}$ by a factor λ , that is

$$\mathbf{F}\mathbf{e} = \mathbf{e} + (\lambda - 1)(\mathbf{e} \cdot \mathbf{e})\mathbf{e} = \lambda\mathbf{e}.$$

This type of deformation is a special case of a stretch deformation introduced earlier. In this case, the body is extended by a factor λ along the direction $\mathbf{e}$ holding $\mathbf{Y}$ fixed, but no deformation occurs in directions orthogonal to $\mathbf{e}$. Moreover, there is no change in the overall orientation of the body, as illustrated in Figure 4.5. Notice that when $\lambda = 1$ we have $\mathbf{F} = \mathbf{I}$, which yields $\varphi(\mathbf{X}) = \mathbf{X}$. In this case, each point of the body is fixed and the body is unchanged.

The following result shows that the two stretches $\mathbf{s}_1$ and $\mathbf{s}_2$ appearing in Result 4.2 can be expressed as the composition of three extensions defined by the eigenvalues and eigenvectors of $\mathbf{U}$ and $\mathbf{V}$.

Result 4.3 Stretches as Extensions. Let $\mathbf{s}_1$ and $\mathbf{s}_2$ be the stretches defined in Result 4.2 and let $\{\lambda_i, \mathbf{u}_i\}$ and $\{\lambda_i, \mathbf{v}_i\}$ ($i = 1, 2, 3$) be the eigenpairs associated with $\mathbf{U}$ and $\mathbf{V}$, respectively. Then

$$\mathbf{s}_1 = \mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3 \quad \text{and} \quad \mathbf{s}_2 = \mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3,$$

where $\mathbf{f}_i$ is an extension at $\mathbf{Y}$ by λ_i in the direction $\mathbf{u}_i$, and $\mathbf{h}_i$ is an extension at $\mathbf{Y}$ by λ_i in the direction $\mathbf{v}_i$. □

Proof See Exercise 11. □

Remarks:

- (1) The tensors $\mathbf{U}$ and $\mathbf{V}$ appearing in the right and left polar decompositions of $\mathbf{F}$ have the same eigenvalues, but different eigenvectors in general (see Exercise 12). Using this fact it follows that the stretches $\mathbf{s}_1$ and $\mathbf{s}_2$ give rise to extensions by the same amounts, but in different directions.
- (2) Motivated by the above result, we call the eigenvalues λ_i the **principal stretches** associated with a deformation gradient $\mathbf{F}$, and call the eigenvectors $\mathbf{u}_i$ and $\mathbf{v}_i$ the right and left **principal directions**. Moreover, we call $\mathbf{U}$ and $\mathbf{V}$ the right and left **stretch tensors**, respectively.
- (3) When a principal stretch $\lambda_i = 1$ for some i , the body is not extended along the corresponding principal direction. Whether it is the right or left principal direction depends on whether the rotation $\mathbf{r}$ is applied last or first, respectively (see Result 4.2 and summary below). When the principal stretches are all unity we have $\mathbf{U} = \mathbf{V} = \mathbf{I}$. In this case, $\mathbf{s}_1(\mathbf{X}) = \mathbf{s}_2(\mathbf{X}) = \mathbf{X}$ and the body is not extended in any direction. □

4.3.2.4 Summary

We have shown that, given any point $\mathbf{Y}$, an arbitrary homogeneous deformation φ can be decomposed as

$$\varphi = \mathbf{d}_1 \circ \mathbf{g} = \mathbf{g} \circ \mathbf{d}_2,$$

where $\mathbf{d}_1$ and $\mathbf{d}_2$ are translations, and $\mathbf{g}$ has a fixed point at $\mathbf{Y}$. The deformation $\mathbf{g}$ can itself be decomposed as

$$\mathbf{g} = \mathbf{r} \circ \mathbf{s}_1 = \mathbf{s}_2 \circ \mathbf{r},$$

where $\mathbf{r}$ is a rotation about $\mathbf{Y}$, and $\mathbf{s}_1$ and $\mathbf{s}_2$ are stretches from $\mathbf{Y}$. Furthermore, these two stretches can be decomposed as

$$\mathbf{s}_1 = \mathbf{f}_1 \circ \mathbf{f}_2 \circ \mathbf{f}_3 \quad \text{and} \quad \mathbf{s}_2 = \mathbf{h}_1 \circ \mathbf{h}_2 \circ \mathbf{h}_3,$$

where $\mathbf{f}_i$ and $\mathbf{h}_i$ are extensions from $\mathbf{Y}$ along the right and left principal directions, respectively. The amount of each extension is determined by the corresponding principal stretch. The principal stretches, principal directions and rotation are entirely determined by the polar decomposition of the deformation gradient $\mathbf{F}$.

From the above results we deduce that an arbitrary homogeneous deformation φ can be understood in several different, but equivalent ways. For example, we may write $\varphi = \mathbf{s}_2 \circ \mathbf{r} \circ \mathbf{d}_2$, which implies that φ can be decomposed into a sequence of three elementary operations: a translation, followed by a rotation, followed by extensions along the left principal directions. Equivalently, we may write $\varphi = \mathbf{r} \circ \mathbf{s}_1 \circ \mathbf{d}_2$, which implies that φ can be decomposed into a different sequence: a translation, followed by extensions along the right principal directions, followed by a rotation (see Exercise 13). Other decompositions are also possible.

The above results about homogeneous deformations can also be applied to general deformations for which the deformation gradient field is not constant. In particular, the Taylor expansion in (4.2) shows that a general deformation is approximately homogeneous in a small neighborhood of each point $\mathbf{X}_0 \in B$. Thus the polar decomposition of the deformation gradient $\mathbf{F}(\mathbf{X}_0)$ completely determines how material in a neighborhood of $\mathbf{X}_0$ is stretched and rotated. For this reason the deformation gradient plays a central role in the modeling of stress in various types of material bodies.

4.3.3 The Cauchy–Green Strain Tensor $\mathbf{C}$

Consider a general deformation $\varphi : B \rightarrow B'$ with associated deformation gradient $\mathbf{F} = \nabla \varphi$. Then another measure of strain is provided by

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}.$$

We call $\mathbf{C} : B \rightarrow \mathcal{V}^2$ the **(right) Cauchy–Green strain tensor** field associated with φ . Notice that $\mathbf{C}$ is symmetric and positive-definite at each point in B .

While $\mathbf{F}$ contains a mixture of information on both rotations and

stretches, $\mathbf{C}$ contains information on stretches only. This observation follows from the right polar decomposition $\mathbf{F} = \mathbf{R}\mathbf{U}$, which implies

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = \mathbf{U}^2.$$

Notice that the rotation tensor $\mathbf{R}$, which is implicit in $\mathbf{F}$, does not appear in $\mathbf{C}$. For this reason, $\mathbf{C}$ is typically a more useful measure of strain than $\mathbf{F}$ (see Exercise 14). The tensor $\mathbf{C}$ will be employed in Chapters 7 and 9 when we study models of stress in elastic and thermoelastic solids at large strains.

Remarks:

- (1) The right stretch tensor $\mathbf{U}$ is also independent of $\mathbf{R}$ and contains information on stretches only. However, we introduce $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ because it is often easier to compute than $\mathbf{U} = \sqrt{\mathbf{F}^T \mathbf{F}}$. In particular, we avoid the tensor square root (see Exercise 15).
- (2) By Result 1.9, the right stretch tensor can be written as $\mathbf{U} = \sum_{i=1}^3 \lambda_i \mathbf{u}_i \otimes \mathbf{u}_i$, where $\lambda_i > 0$ are the principal stretches and $\mathbf{u}_i$ are the right principal directions. From the relation $\mathbf{C} = \mathbf{U}^2$ we deduce $\mathbf{C} = \sum_{i=1}^3 \lambda_i^2 \mathbf{u}_i \otimes \mathbf{u}_i$, which implies that the eigenvalues of $\mathbf{C}$ are the squares of the principal stretches, and the eigenvectors are the right principal directions. Thus the principal stretches and directions can be determined from $\mathbf{C}$ without calculating $\mathbf{U}$.
- (3) Another measure of strain related to $\mathbf{C}$ is the **left Cauchy–Green strain tensor** $\mathbf{B} = \mathbf{F}\mathbf{F}^T$. However, this strain tensor will not be employed in any of our developments. $\square$

4.3.4 Interpretation of $\mathbf{C}$

Here we show how changes in the relative position and orientation of points in a material body are quantified by $\mathbf{C}$. To begin, consider an arbitrary point $\mathbf{X}$ in the reference configuration B , and let Ω be the open ball of radius $\alpha > 0$ centered at $\mathbf{X}$. For any two unit vectors $\mathbf{e}$ and $\mathbf{d}$ consider the points $\mathbf{Y} = \mathbf{X} + \alpha\mathbf{e}$ and $\mathbf{Z} = \mathbf{X} + \alpha\mathbf{d}$ as shown in Figure 4.6. Let $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ denote the corresponding points in Ω' , and let $\phi \in [0, \pi]$ denote the angle between the vectors $\mathbf{v} = \mathbf{y} - \mathbf{x}$ and $\mathbf{w} = \mathbf{z} - \mathbf{x}$.

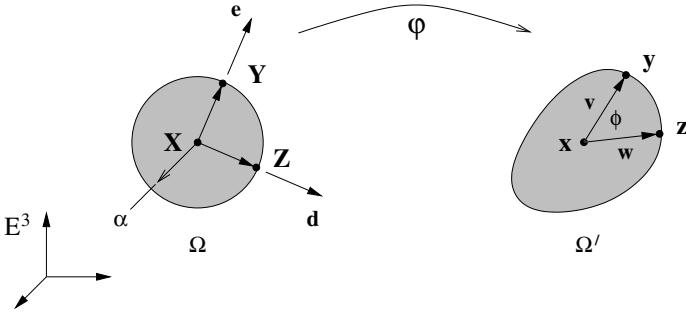


Fig. 4.6 Setup for interpreting the strain tensor $\mathbf{C}$. Three points $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ in Ω are mapped to corresponding points $\mathbf{x}, \mathbf{y}, \mathbf{z}$ in Ω' . The change in the relative position and orientation of the points can be quantified in terms of $\mathbf{C}$.

Result 4.4 Cauchy–Green Strain Relations. For any point $\mathbf{X} \in B$ and unit vectors $\mathbf{e}$ and $\mathbf{d}$ define $\lambda(\mathbf{e}) > 0$ and $\theta(\mathbf{e}, \mathbf{d}) \in [0, \pi]$ by

$$\lambda(\mathbf{e}) = \sqrt{\mathbf{e} \cdot \mathbf{C} \mathbf{e}} \quad \text{and} \quad \cos \theta(\mathbf{e}, \mathbf{d}) = \frac{\mathbf{e} \cdot \mathbf{C} \mathbf{d}}{\sqrt{\mathbf{e} \cdot \mathbf{C} \mathbf{e}} \sqrt{\mathbf{d} \cdot \mathbf{C} \mathbf{d}}}.$$

Then as $\alpha \rightarrow 0$ we have

$$\frac{|\mathbf{y} - \mathbf{x}|}{|\mathbf{Y} - \mathbf{X}|} \rightarrow \lambda(\mathbf{e}), \quad \frac{|\mathbf{z} - \mathbf{x}|}{|\mathbf{Z} - \mathbf{X}|} \rightarrow \lambda(\mathbf{d}) \quad (4.5)$$

and

$$\cos \phi \rightarrow \cos \theta(\mathbf{e}, \mathbf{d}). \quad (4.6)$$

□

Proof To establish (4.5)₁ we notice that $\mathbf{v} = \mathbf{y} - \mathbf{x}$ can be written as

$$\mathbf{v} = \varphi(\mathbf{Y}) - \varphi(\mathbf{X}) = \varphi(\mathbf{X} + \alpha \mathbf{e}) - \varphi(\mathbf{X}).$$

Using a Taylor expansion about $\alpha = 0$ we obtain

$$\mathbf{v} = \alpha \mathbf{F}(\mathbf{X}) \mathbf{e} + \mathcal{O}(\alpha^2), \quad (4.7)$$

and using the relations $|\mathbf{v}|^2 = \mathbf{v} \cdot \mathbf{v}$ and $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ we find

$$|\mathbf{v}|^2 = \alpha^2 \mathbf{F}(\mathbf{X}) \mathbf{e} \cdot \mathbf{F}(\mathbf{X}) \mathbf{e} + \mathcal{O}(\alpha^3) = \alpha^2 \mathbf{e} \cdot \mathbf{C}(\mathbf{X}) \mathbf{e} + \mathcal{O}(\alpha^3).$$

Dividing through by α^2 and noting that $|\mathbf{y} - \mathbf{x}| = |\mathbf{v}|$ and $|\mathbf{Y} - \mathbf{X}| = \alpha$ we get

$$\frac{|\mathbf{y} - \mathbf{x}|^2}{|\mathbf{Y} - \mathbf{X}|^2} = \frac{|\mathbf{v}|^2}{\alpha^2} = \mathbf{e} \cdot \mathbf{C}(\mathbf{X})\mathbf{e} + \mathcal{O}(\alpha), \quad (4.8)$$

which leads to (4.5)₁. To establish (4.5)₂ we notice that $\mathbf{w} = \mathbf{z} - \mathbf{x}$ can be written as

$$\mathbf{w} = \varphi(\mathbf{Z}) - \varphi(\mathbf{X}) = \varphi(\mathbf{X} + \alpha\mathbf{d}) - \varphi(\mathbf{X}).$$

Using a Taylor expansion as above we obtain

$$\mathbf{w} = \alpha\mathbf{F}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha^2), \quad (4.9)$$

which implies

$$|\mathbf{w}|^2 = \alpha^2 \mathbf{d} \cdot \mathbf{C}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha^3).$$

Dividing through by α^2 and noting that $|\mathbf{z} - \mathbf{x}| = |\mathbf{w}|$ and $|\mathbf{Z} - \mathbf{X}| = \alpha$ gives

$$\frac{|\mathbf{z} - \mathbf{x}|^2}{|\mathbf{Z} - \mathbf{X}|^2} = \frac{|\mathbf{w}|^2}{\alpha^2} = \mathbf{d} \cdot \mathbf{C}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha), \quad (4.10)$$

which leads to (4.5)₂. To establish (4.6) we use the relation

$$\cos \phi = \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}||\mathbf{w}|}. \quad (4.11)$$

From (4.7) and (4.9) we get

$$\mathbf{v} \cdot \mathbf{w} = \alpha^2 \mathbf{F}(\mathbf{X})\mathbf{e} \cdot \mathbf{F}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha^3) = \alpha^2 \mathbf{e} \cdot \mathbf{C}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha^3),$$

which implies

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\alpha^2} = \mathbf{e} \cdot \mathbf{C}(\mathbf{X})\mathbf{d} + \mathcal{O}(\alpha). \quad (4.12)$$

When (4.12), (4.10) and (4.8) are combined with (4.11) we obtain the result in (4.6). $\square$

Remarks:

- (1) The above result shows that $\lambda(\mathbf{e})$ is the limiting value of the ratio $|\mathbf{y} - \mathbf{x}|/|\mathbf{Y} - \mathbf{X}|$ as $\mathbf{Y}$ tends to $\mathbf{X}$ along a direction $\mathbf{e}$. For this reason $\lambda(\mathbf{e})$ is called the **stretch** in the direction $\mathbf{e}$ at $\mathbf{X}$. It is the ratio of deformed length to initial length of an infinitesimal line segment that, prior to deformation, was in the direction $\mathbf{e}$ at $\mathbf{X}$ (see Exercise 16).

- (2) Let $\mathbf{u}_i$ be a right principal direction at $\mathbf{X}$ with corresponding principal stretch λ_i so that $\mathbf{C}\mathbf{u}_i = \lambda_i^2 \mathbf{u}_i$ (no sum). Then a straightforward calculation shows that $\lambda(\mathbf{u}_i) = \lambda_i$, which provides further justification for referring to λ_i ($i = 1, 2, 3$) as the principal stretches.
- (3) An analysis similar to that in Section 3.5.3 reveals that, at any point $\mathbf{X}$, the extreme values of $\lambda(\mathbf{e})$ occur when $\mathbf{e}$ is an eigenvector of $\mathbf{C}$ (right principal direction). From this we deduce that the extreme values of $\lambda(\mathbf{e})$ at a point are given by the maximum and minimum principle stretches at that point.
- (4) The angle $\theta(\mathbf{e}, \mathbf{d})$ is the limiting value of ϕ as $\mathbf{Y}, \mathbf{Z}$ tend to $\mathbf{X}$ along directions $\mathbf{e}, \mathbf{d}$. If we denote the angle between $\mathbf{e}$ and $\mathbf{d}$ by $\Theta(\mathbf{e}, \mathbf{d})$, then the quantity $\gamma(\mathbf{e}, \mathbf{d}) = \Theta(\mathbf{e}, \mathbf{d}) - \theta(\mathbf{e}, \mathbf{d})$ is called the **shear** between the directions $\mathbf{e}$ and $\mathbf{d}$ at $\mathbf{X}$. Physically, shear is the change in angle between two infinitesimal line segments that, prior to deformation, were aligned with $\mathbf{e}$ and $\mathbf{d}$ at $\mathbf{X}$. A straightforward analysis shows that the shear between any two eigenvectors of $\mathbf{C}$ (right principal directions) is zero (see Exercises 17 and 18). $\square$

The following result shows that, in contrast to $\mathbf{F}$, the components of $\mathbf{C}$ explicitly quantify the stretch and shear caused by a deformation φ .

Result 4.5 Components of $\mathbf{C}$. *Let C_{ij} be the components of $\mathbf{C}$ in an arbitrary coordinate frame $\{\mathbf{e}_i\}$. Then for any point $\mathbf{X} \in B$ we have*

$$C_{ii} = \lambda^2(\mathbf{e}_i) \quad \text{and} \quad C_{ij} = \lambda(\mathbf{e}_i)\lambda(\mathbf{e}_j) \sin \gamma(\mathbf{e}_i, \mathbf{e}_j) \quad (\text{no sum, } i \neq j).$$

Thus the diagonal components of $\mathbf{C}$ are the squares of the stretches along the coordinate directions, and the off-diagonal components are related to the shear between the corresponding pairs of coordinate directions. $\square$

Proof See Exercise 19. $\square$

4.3.5 Rigid Deformations

A (homogeneous) deformation $\varphi : B \rightarrow B'$ is called **rigid** if

$$\varphi(\mathbf{X}) = \mathbf{c} + \mathbf{Q}\mathbf{X}$$

for some vector $\mathbf{c}$ and rotation tensor $\mathbf{Q}$. Notice that this class of deformations contains both translations and rotations as considered in Section 4.3.2. For a rigid deformation we have $\mathbf{F} = \mathbf{Q}$, which yields

$$\mathbf{C}(\mathbf{X}) = \mathbf{F}(\mathbf{X})^T \mathbf{F}(\mathbf{X}) = \mathbf{Q}^T \mathbf{Q} = \mathbf{I}, \quad \forall \mathbf{X} \in B.$$

In view of Results 4.4 and 4.5 we see that rigid deformations produce no strain as measured by $\mathbf{C}$. In particular, the stretch in every direction is unity and the shear between any two directions is zero at each point in B . Indeed, an inspection of the proof of Result 4.4 reveals that the relative position and orientation between *any* three points in B is unchanged by a rigid deformation. Rigid deformations can be completely characterized in terms of the tensor field $\mathbf{C}$. In particular, it can be shown that a deformation φ is rigid if and only if $\mathbf{C}(\mathbf{X}) = \mathbf{I}$ for all $\mathbf{X} \in B$.

4.3.6 The Infinitesimal Strain Tensor $\mathbf{E}$

Consider a deformation $\varphi : B \rightarrow B'$ with associated displacement field $\mathbf{u}$ and displacement gradient $\nabla \mathbf{u}$. Then another measure of strain is provided by

$$\mathbf{E} = \text{sym}(\nabla \mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T).$$

We call $\mathbf{E} : B \rightarrow \mathcal{V}^2$ the **infinitesimal strain tensor** field associated with φ . Notice that, by definition, $\mathbf{E}$ is symmetric at each point in B .

The tensor $\mathbf{E}$ is related to the deformation gradient $\mathbf{F}$ and the Cauchy-Green tensor $\mathbf{C}$. In particular, from the definition of $\mathbf{u}$ in (4.1) we deduce that $\nabla \mathbf{u} = \mathbf{F} - \mathbf{I}$, which leads to the result

$$\mathbf{E} = \text{sym}(\mathbf{F} - \mathbf{I}).$$

Moreover, since $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ we also find

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}) - \frac{1}{2}\nabla \mathbf{u}^T \nabla \mathbf{u}. \quad (4.13)$$

The tensor $\mathbf{E}$ is particularly useful in the case of small deformations. We say that a deformation φ is **small** if there is a number $0 \leq \epsilon \ll 1$ such that $|\nabla \mathbf{u}| = \mathcal{O}(\epsilon)$, or equivalently $\partial u_i / \partial X_j = \mathcal{O}(\epsilon)$, for all points $\mathbf{X} \in B$. Thus a deformation is small if the norm of the displacement gradient is small at all points in the body. In this case we deduce from (4.13) that

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}) + \mathcal{O}(\epsilon^2). \quad (4.14)$$

Notice that, if terms of order $\mathcal{O}(\epsilon^2)$ are neglected, then $\mathbf{E}$ is equivalent

to $\mathbf{C}$ up to a constant multiplicative factor and offset. The tensor $\mathbf{E}$ will arise naturally in Chapters 7 and 9 when we study linearized models of stress in elastic and thermoelastic solids.

Remarks:

- (1) When $\nabla \mathbf{u} = \mathbf{O}$ for all $\mathbf{X} \in B$ we have $\mathbf{F} = \mathbf{I}$ for all $\mathbf{X} \in B$, which implies that $\boldsymbol{\varphi}$ is a translation. Thus $\boldsymbol{\varphi}$ is a small deformation when it deviates only slightly from a pure translation (see Exercise 20).
- (2) For small deformations the tensor $\mathbf{E}$ contains essentially the same information as $\mathbf{C}$. However, there is one important difference: $\mathbf{E}$ depends linearly on $\mathbf{u}$ (hence $\boldsymbol{\varphi}$), whereas $\mathbf{C}$ depends non-linearly on $\mathbf{u}$. □

4.3.7 Interpretation of $\mathbf{E}$

Here we describe the way in which $\mathbf{E}$ provides a measure of strain for a small deformation. In particular, we consider a deformation $\boldsymbol{\varphi}$ and assume it is small in the sense that $\partial u_i / \partial X_j = \mathcal{O}(\epsilon)$ for all $\mathbf{X} \in B$ where $0 \leq \epsilon \ll 1$.

Result 4.6 Components of $\mathbf{E}$. *Let E_{ij} be the components of $\mathbf{E}$ in an arbitrary coordinate frame $\{\mathbf{e}_i\}$. If we neglect terms of order $\mathcal{O}(\epsilon^2)$, then for any point $\mathbf{X} \in B$ we have*

$$E_{ii} \approx \lambda(\mathbf{e}_i) - 1 \quad \text{and} \quad E_{ij} \approx \frac{1}{2} \sin \gamma(\mathbf{e}_i, \mathbf{e}_j) \quad (\text{no sum, } i \neq j),$$

where $\lambda(\mathbf{e}_i)$ is the stretch in the direction $\mathbf{e}_i$ and $\gamma(\mathbf{e}_i, \mathbf{e}_j)$ is the shear between the directions $\mathbf{e}_i$ and $\mathbf{e}_j$. □

Proof To establish the result for the diagonal components we consider the case $i = 1$. From (4.14) we have

$$C_{11} = 1 + 2E_{11} + \mathcal{O}(\epsilon^2),$$

and since $E_{11} = \mathcal{O}(\epsilon)$ it follows by properties of the square root function that

$$\sqrt{C_{11}} = 1 + E_{11} + \mathcal{O}(\epsilon^2).$$

Neglecting the terms of order $\mathcal{O}(\epsilon^2)$ and using Result 4.5 gives

$$E_{11} \approx \sqrt{C_{11}} - 1 = \lambda(\mathbf{e}_1) - 1,$$

which is the desired result. The other diagonal cases follow similarly. To establish the result for the off-diagonal components we consider the case $i = 1$ and $j = 2$. Then from Result 4.5 we have

$$\sin \gamma(\mathbf{e}_1, \mathbf{e}_2) = \frac{C_{12}}{\sqrt{C_{11}}\sqrt{C_{22}}}. \quad (4.15)$$

From (4.14) and the fact that E_{11} , E_{22} and E_{12} are all order $\mathcal{O}(\epsilon)$ we obtain

$$C_{12} = 2E_{12} + \mathcal{O}(\epsilon^2), \quad C_{11} = 1 + \mathcal{O}(\epsilon), \quad C_{22} = 1 + \mathcal{O}(\epsilon),$$

which together with (4.15) implies

$$\sin \gamma(\mathbf{e}_1, \mathbf{e}_2) = 2E_{12} + \mathcal{O}(\epsilon^2).$$

Neglecting the terms of order $\mathcal{O}(\epsilon^2)$ gives

$$E_{12} \approx \frac{1}{2} \sin \gamma(\mathbf{e}_1, \mathbf{e}_2),$$

which is the desired result. The other off-diagonal cases follow similarly. $\square$

Remarks:

- (1) As shown in Result 4.4, the stretch $\lambda(\mathbf{e}_i)$ is the limiting value of the ratio $|\mathbf{y} - \mathbf{x}|/|\mathbf{Y} - \mathbf{X}|$ as $\mathbf{Y}$ tends to $\mathbf{X}$ along a direction $\mathbf{e}_i$ (see Figure 4.6). From this we deduce that $\lambda(\mathbf{e}_i) - 1$ is the limiting value of the quantity

$$\frac{|\mathbf{y} - \mathbf{x}|}{|\mathbf{Y} - \mathbf{X}|} - 1 = \frac{|\mathbf{y} - \mathbf{x}| - |\mathbf{Y} - \mathbf{X}|}{|\mathbf{Y} - \mathbf{X}|}.$$

Thus, for a small deformation, the diagonal component E_{ii} (no sum) is approximately equal to the relative change in length of an infinitesimal line segment that, prior to deformation, was in the direction $\mathbf{e}_i$ at $\mathbf{X}$.

- (2) When the shear angle $\gamma(\mathbf{e}_i, \mathbf{e}_j)$ is small we obtain

$$E_{ij} \approx \frac{1}{2} \sin \gamma(\mathbf{e}_i, \mathbf{e}_j) \approx \frac{1}{2} \gamma(\mathbf{e}_i, \mathbf{e}_j).$$

Thus, for a small deformation, the off-diagonal component E_{ij} is approximately equal to half of the shear angle between two infinitesimal line segments that, prior to deformation, were in the directions $\mathbf{e}_i$ and $\mathbf{e}_j$ at $\mathbf{X}$ (see Exercise 21). $\square$

4.3.8 Infinitesimally Rigid Deformations

A (homogeneous) deformation $\varphi : B \rightarrow B'$ is called **infinitesimally rigid** if the associated displacement field $\mathbf{u}$ is of the form

$$\mathbf{u}(\mathbf{X}) = \mathbf{c} + \mathbf{W}\mathbf{X}, \quad \forall \mathbf{X} \in B$$

for some vector $\mathbf{c}$ and skew-symmetric tensor $\mathbf{W}$. Equivalently, by Result 1.3, the displacement field for an infinitesimally rigid deformation may be written as

$$\mathbf{u}(\mathbf{X}) = \mathbf{c} + \mathbf{w} \times \mathbf{X},$$

where $\mathbf{w}$ is the axial vector of $\mathbf{W}$. As the name suggests, an infinitesimally rigid deformation is related to a small rigid deformation (see Exercise 22). For an infinitesimally rigid deformation the displacement gradient is $\nabla \mathbf{u} = \mathbf{W}$ and the infinitesimal strain tensor is given by

$$\mathbf{E}(\mathbf{X}) = \text{sym } \nabla \mathbf{u}(\mathbf{X}) = \frac{1}{2}(\mathbf{W} + \mathbf{W}^T) = \mathbf{O}, \quad \forall \mathbf{X} \in B.$$

In view of Result 4.6, we see that an infinitesimally rigid deformation produces no strain as measured by $\mathbf{E}$. In particular, the change in length of an infinitesimal line segment along any coordinate direction is approximately zero, and the shear between any two coordinate directions is approximately zero at each point in B . Infinitesimally rigid deformations can be completely characterized in terms of the tensor field $\mathbf{E}$. In particular, it can be shown that a deformation φ is infinitesimally rigid if and only if $\mathbf{E}(\mathbf{X}) = \mathbf{O}$ for all $\mathbf{X} \in B$.

4.4 Motions

The continuous deformation of a body over the course of time is called a **motion**. The motion of a body with reference configuration B is described by a continuous map $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$, where for each fixed $t \geq 0$ the function $\varphi(\cdot, t) = \varphi_t : B \rightarrow \mathbb{E}^3$ is a deformation of B (see Exercise 23). At any time $t \geq 0$ the deformation φ_t maps the reference configuration B onto a configuration $B_t = \varphi_t(B)$. We call B_t the current or deformed configuration at time t (see Figure 4.7).

We assume that φ_0 is the identity map in the sense that $\varphi_0(\mathbf{X}) = \mathbf{X}$ for all $\mathbf{X} \in B$, which implies $B_0 = B$. Thus a motion represents a continuous deformation of a body that begins from the configuration B . The assumption of continuity implies that, during a motion, a body cannot break apart into disjoint pieces, develop holes and so on. We further assume that for each $t \geq 0$ the deformation $\varphi_t : B \rightarrow B_t$ is

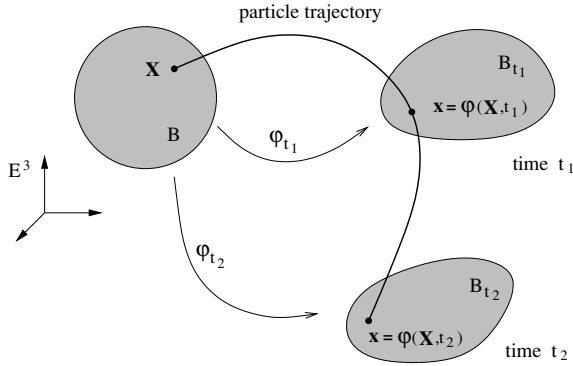


Fig. 4.7 Various configurations in the motion of a material body: B denotes the reference configuration at time $t = 0$, and B_t denotes the current configuration at time t .

admissible as described in Section 4.2. Thus we can define an inverse deformation $\psi_t = \varphi_t^{-1} : B_t \rightarrow B$ such that

$$\mathbf{X} = \psi_t(\mathbf{x}) = \psi(\mathbf{x}, t).$$

By properties of inverse functions, for any $t \geq 0$ we have

$$\mathbf{X} = \psi_t(\varphi_t(\mathbf{X})) = \psi(\varphi(\mathbf{X}, t), t) \quad \forall \mathbf{X} \in B,$$

and similarly

$$\mathbf{x} = \varphi_t(\psi_t(\mathbf{x})) = \varphi(\psi(\mathbf{x}, t), t) \quad \forall \mathbf{x} \in B_t$$

(see Exercise 24).

Throughout the remainder of our developments we assume that all motions φ and their inverses ψ are smooth in the sense that partial derivatives of all orders exist and are continuous. This is always stronger than what we need, but allows for a clean presentation.

4.4.1 Material and Spatial Fields

In our study of the motion of continuum bodies we will encounter fields defined over the current configuration B_t whose points we label by $\mathbf{x}$. However, since $\mathbf{x} = \varphi(\mathbf{X}, t)$, any function of $\mathbf{x} \in B_t$ can also be expressed as a function of $\mathbf{X} \in B$. Similarly, we will encounter fields defined over the reference configuration B whose points we label by $\mathbf{X}$. However, since $\mathbf{X} = \psi(\mathbf{x}, t)$, any function of $\mathbf{X} \in B$ can also be expressed as a function of $\mathbf{x} \in B_t$. To help keep track of where a field was

originally defined, and how it is currently being expressed, we introduce the following definitions.

By a **material field** we mean a field expressed in terms of the points $\mathbf{X}$ of B ; for example, $\Omega = \Omega(\mathbf{X}, t)$. By a **spatial field** we mean a field expressed in terms of the points $\mathbf{x}$ of B_t ; for example, $\Gamma = \Gamma(\mathbf{x}, t)$. To any material field $\Omega(\mathbf{X}, t)$ we can associate a spatial field $\Omega_s(\mathbf{x}, t)$ by the relation

$$\Omega_s(\mathbf{x}, t) = \Omega(\boldsymbol{\psi}(\mathbf{x}, t), t).$$

We call Ω_s the **spatial description** of the material field Ω . Similarly, to any spatial field $\Gamma(\mathbf{x}, t)$ we can associate a material field $\Gamma_m(\mathbf{X}, t)$ by the relation

$$\Gamma_m(\mathbf{X}, t) = \Gamma(\boldsymbol{\varphi}(\mathbf{X}, t), t).$$

We call Γ_m the **material description** of the spatial field Γ (see Exercise 25).

4.4.2 Coordinate Derivatives

Throughout the remainder of our studies we will need to be careful about distinguishing between material coordinates $\mathbf{X} = (X_1, X_2, X_3)^T$, which label points in B , and spatial coordinates $\mathbf{x} = (x_1, x_2, x_3)^T$, which label points in the current configuration B_t . To distinguish between derivatives with respect to these two sets of coordinates we introduce the following notation.

We use the symbol ∇^x to denote the gradient, divergence and curl of material fields with respect to the material coordinates X_i for any fixed time $t \geq 0$. Similarly, we use the symbol ∇^x to denote the gradient, divergence and curl of spatial fields with respect to the spatial coordinates x_i for any fixed time $t \geq 0$. By extension we define the Laplacian operators $\Delta^x = \nabla^x \cdot (\nabla^x)$ and $\Delta^x = \nabla^x \cdot (\nabla^x)$.

4.4.3 Time Derivatives

In our studies we will frequently need to compute the rate of change of a given field with respect to time. By the **total time derivative** of a field we mean the rate of change of the field as measured by an observer who is tracking the motion of each particle in the body. It is important to keep in mind that, because the reference configuration B is fixed, the material coordinates $\mathbf{X}$ of each particle are fixed. In contrast, because

the current configuration B_t changes with time, the spatial coordinates $\mathbf{x}$ of each particle change with time. The precise manner in which the spatial coordinates change with time is given by the motion, namely

$$\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t).$$

Using an overdot to denote the total time derivative, we note from the above remarks that the total time derivative of a material field $\Omega(\mathbf{X}, t)$ is simply

$$\dot{\Omega}(\mathbf{X}, t) = \frac{\partial}{\partial t} \Omega(\mathbf{X}, t).$$

On the other hand, because $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$, the total time derivative of a spatial field $\Gamma(\mathbf{x}, t)$ is given by

$$\dot{\Gamma}(\mathbf{x}, t) = \left[\frac{\partial}{\partial t} \Gamma(\boldsymbol{\varphi}(\mathbf{X}, t), t) \right] \Big|_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)} = \left[\dot{\Gamma}_m(\mathbf{X}, t) \right] \Big|_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)}, \quad (4.16)$$

that is

$$\dot{\Gamma} = [\dot{\Gamma}_m]_s.$$

Remarks:

- (1) The total time derivative of a field is often called the **material** or **substantial**, or sometimes **convective**, time derivative. It is the rate of change of the field that we would measure if we were to follow each material particle as it moves in space. Because each particle is identified with fixed material coordinates $\mathbf{X}$, it is the time derivative computed with $\mathbf{X}$ fixed.
- (2) In general, notice that $\dot{\Gamma}(\mathbf{x}, t) \neq \frac{\partial}{\partial t} \Gamma(\mathbf{x}, t)$. That is, the total time derivative of a spatial field is different from the partial time derivative computed with $\mathbf{x}$ fixed. In this case the partial time derivative fails to account for the fact that the spatial coordinates $\mathbf{x}$ of a body particle change with time, namely $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ (see Exercise 26). □

4.4.4 Velocity and Acceleration Fields

Let $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body and consider a material particle labeled by $\mathbf{X}$ in the reference configuration B . At any time $t \geq 0$, this particle is labeled by $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ in the current configuration B_t (see Figure 4.7). We denote by $\mathbf{V}(\mathbf{X}, t)$ the **velocity**

at time t of the material particle labeled by $\mathbf{X}$ in B . By definition of the motion we have

$$\mathbf{V}(\mathbf{X}, t) = \frac{\partial}{\partial t} \boldsymbol{\varphi}(\mathbf{X}, t).$$

Similarly, we denote by $\mathbf{A}(\mathbf{X}, t)$ the **acceleration** at time t of the material particle labeled by $\mathbf{X}$ in B . By definition of the motion we have

$$\mathbf{A}(\mathbf{X}, t) = \frac{\partial^2}{\partial t^2} \boldsymbol{\varphi}(\mathbf{X}, t).$$

From the above definitions we see that the velocity and acceleration of material particles are naturally material fields. Frequently, however, we will need the spatial descriptions of these fields. We denote by $\mathbf{v}(\mathbf{x}, t)$ the spatial description of the material velocity field $\mathbf{V}(\mathbf{X}, t)$, that is

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}_s(\mathbf{x}, t) = \left[\frac{\partial}{\partial t} \boldsymbol{\varphi}(\mathbf{X}, t) \right] \Big|_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)}, \quad (4.17)$$

and we denote by $\mathbf{a}(\mathbf{x}, t)$ the spatial description of the material acceleration field $\mathbf{A}(\mathbf{X}, t)$, so that

$$\mathbf{a}(\mathbf{x}, t) = \mathbf{A}_s(\mathbf{x}, t) = \left[\frac{\partial^2}{\partial t^2} \boldsymbol{\varphi}(\mathbf{X}, t) \right] \Big|_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)}$$

(see Exercises 27 and 28). Notice that $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{a}(\mathbf{x}, t)$ correspond to the velocity and acceleration of the material particle whose current coordinates are $\mathbf{x}$ at time t . The following result shows that the spatial velocity field can be used to derive a convenient formula for the total time derivative of an arbitrary spatial field.

Result 4.7 Total Time Derivative. *Let $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}$, and consider an arbitrary spatial scalar field $\phi = \phi(\mathbf{x}, t)$ and an arbitrary spatial vector field $\mathbf{w} = \mathbf{w}(\mathbf{x}, t)$. Then the total time derivatives of ϕ and $\mathbf{w}$ are given by*

$$\dot{\phi} = \frac{\partial}{\partial t} \phi + \nabla^x \phi \cdot \mathbf{v} \quad \text{and} \quad \dot{\mathbf{w}} = \frac{\partial}{\partial t} \mathbf{w} + (\nabla^x \mathbf{w}) \mathbf{v}.$$

□

Proof By definition of the gradients of ϕ and $\mathbf{w}$ we have

$$\nabla^x \phi = \frac{\partial \phi}{\partial x_i} \mathbf{e}_i \quad \text{and} \quad \nabla^x \mathbf{w} = \frac{\partial w_i}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j,$$

where $\{\mathbf{e}_k\}$ is any fixed frame, and by definition of $\mathbf{v}$ in (4.17) we have

$$\mathbf{v}(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} = \frac{\partial}{\partial t} \boldsymbol{\varphi}(\mathbf{X}, t).$$

Using the definition of the total time derivative in (4.16) we obtain

$$\begin{aligned} & \left[\dot{\phi}(\mathbf{x}, t) \right] \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \\ &= \frac{\partial}{\partial t} \phi(\boldsymbol{\varphi}(\mathbf{X}, t), t) \\ &= \left[\frac{\partial}{\partial t} \phi(\mathbf{x}, t) \right] \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} + \left[\frac{\partial}{\partial x_i} \phi(\mathbf{x}, t) \right] \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \frac{\partial}{\partial t} \varphi_i(\mathbf{X}, t) \\ &= \left[\frac{\partial}{\partial t} \phi(\mathbf{x}, t) \right] \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} + \left[\frac{\partial}{\partial x_i} \phi(\mathbf{x}, t) \right] \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} v_i(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)}. \end{aligned}$$

Expressing the above result in terms of spatial coordinates $\mathbf{x}$ we have

$$\dot{\phi}(\mathbf{x}, t) = \frac{\partial \phi}{\partial t}(\mathbf{x}, t) + \frac{\partial \phi}{\partial x_i}(\mathbf{x}, t) v_i(\mathbf{x}, t),$$

which establishes the result for ϕ . Given $\mathbf{w} = w_i \mathbf{e}_i$ we apply the above result to each scalar field $w_i(\mathbf{x}, t)$ ($i = 1, 2, 3$) to obtain

$$\dot{w}_i(\mathbf{x}, t) = \frac{\partial w_i}{\partial t}(\mathbf{x}, t) + \frac{\partial w_i}{\partial x_j}(\mathbf{x}, t) v_j(\mathbf{x}, t),$$

which implies the result for $\mathbf{w}$. □

Remarks:

- (1) The above result shows that, if the spatial velocity $\mathbf{v}$ is known, then the total time derivative of a spatial field ϕ can be computed without explicit knowledge of the motion $\boldsymbol{\varphi}$ or its inverse $\boldsymbol{\psi}$ (see Exercise 29).
- (2) By definition, the spatial acceleration field $\mathbf{a}$ of a motion satisfies $\mathbf{a} = \dot{\mathbf{v}}$, where $\mathbf{v}$ is the spatial velocity field of the motion. From Result 4.7 we deduce

$$\mathbf{a} = \frac{\partial \mathbf{v}}{\partial t} + (\nabla^x \mathbf{v}) \mathbf{v}.$$

Thus the spatial acceleration is actually a nonlinear function of the spatial velocity and its derivatives.

- (3) Many texts employ the notation $\mathbf{v} \cdot \nabla^x \mathbf{w}$ in place of $(\nabla^x \mathbf{w})\mathbf{v}$. We, however, will always use $(\nabla^x \mathbf{w})\mathbf{v}$, which denotes the usual product between the second-order tensor $\nabla^x \mathbf{w}$ and the vector $\mathbf{v}$.

□

4.5 Rate of Strain, Spin

Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body, and let B_t and $B_{t'}$ denote the configurations at times $t \geq 0$ and $t' \geq t$. Moreover, let Ω be a small ball of radius $\alpha > 0$ centered at a point $\mathbf{x}$ in B_t , and let Ω' be the corresponding region in $B_{t'}$ as depicted in Figure 4.8. Any quantitative measure of the rate of change of shape between Ω' and Ω in the limits $\alpha \rightarrow 0$ and $t' \rightarrow t$ is generally called **rate of strain** at $\mathbf{x}$ and t . Similarly, any quantitative measure of the rate of change of orientation of Ω' with respect to Ω is called **rate of rotation** or **spin**. Notice that, in contrast to the concept of strain, the concepts of rate of strain and spin are independent of the reference configuration B . For this reason rate of strain and spin play a central role in the study of fluids. In the following sections we introduce measures of rate of strain and spin and discuss some of their properties.

4.5.1 Rate of Strain Tensor $\mathbf{L}$ and Spin Tensor $\mathbf{W}$

Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with spatial velocity field $\mathbf{v}$. Then a measure of rate of strain is provided by the tensor

$$\mathbf{L} = \text{sym}(\nabla^x \mathbf{v}) = \frac{1}{2}(\nabla^x \mathbf{v} + \nabla^x \mathbf{v}^T).$$

We call $\mathbf{L}_t = \mathbf{L}(\cdot, t) : B_t \rightarrow \mathcal{V}^2$ the **rate of strain tensor** field associated with $\mathbf{v}$. Notice that, by definition, $\mathbf{L}$ is a spatial field and is symmetric for each point $\mathbf{x}$ in B_t and time t .

To any spatial velocity field $\mathbf{v}$ we associate another tensor field defined by

$$\mathbf{W} = \text{skew}(\nabla^x \mathbf{v}) = \frac{1}{2}(\nabla^x \mathbf{v} - \nabla^x \mathbf{v}^T).$$

We call $\mathbf{W}_t = \mathbf{W}(\cdot, t) : B_t \rightarrow \mathcal{V}^2$ the **spin tensor** field associated with $\mathbf{v}$. By definition, $\mathbf{W}$ is a spatial field and is skew-symmetric for each point $\mathbf{x}$ in B_t and time t .

Whereas $\mathbf{L}$ is a measure of rate of strain or distortion in a material, $\mathbf{W}$ is a measure of rate of rotation or spin (see Exercise 30). These

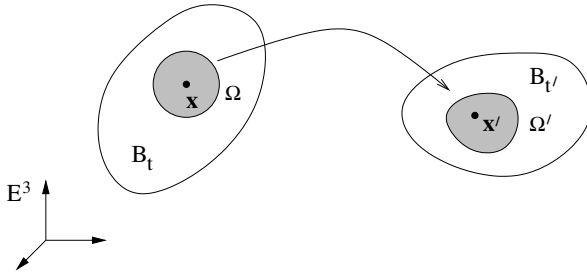


Fig. 4.8 Two configurations B_t and $B_{t'}$ in the motion of a body. The rate of change of shape of a small region Ω in B_t leads to the concept of rate of strain.

tensor fields will arise in Chapters 6 and 8 when we consider theories describing the flow of various types of fluids.

4.5.2 Interpretation of $\mathbf{L}$ and $\mathbf{W}$

By arguments analogous to those used for the infinitesimal strain tensor, we find that the diagonal components of $\mathbf{L}$ quantify the *instantaneous rate of stretching* of infinitesimal line segments located at $\mathbf{x}$ in B_t which are aligned with the coordinate axes. Similarly, the off-diagonal components of $\mathbf{L}$ quantify the *instantaneous rate of shearing*. In particular, if we let $\hat{\mathbf{E}}$ be the infinitesimal strain tensor associated with the small deformation $\hat{\boldsymbol{\varphi}} : B_t \rightarrow B_{t+s}$ ($0 \leq s \ll 1$), then $\mathbf{L}$ is the rate of change of $\hat{\mathbf{E}}$ with respect to s at $s = 0$. Equivalently, if we let $\hat{\mathbf{U}}$ be the right stretch tensor associated with $\hat{\boldsymbol{\varphi}}$, then $\mathbf{L}$ is the rate of change of $\hat{\mathbf{U}}$ with respect to s at $s = 0$.

The skew-symmetric tensor $\mathbf{W}$ quantifies the *instantaneous rate of rigid rotation* of an infinitesimal volume of material at $\mathbf{x}$ in B_t . In particular, if we let $\hat{\mathbf{R}}$ be the rotation tensor in the polar decomposition of the deformation gradient $\nabla^x \hat{\boldsymbol{\varphi}}$, then $\mathbf{W}$ is the rate of change of $\hat{\mathbf{R}}$ with respect to s at $s = 0$ (see Exercise 31).

4.5.3 Vorticity

Let $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow E^3$ be a motion of a continuum body with spatial velocity field $\mathbf{v}$. By the **vorticity** of a motion we mean the spatial vector field

$$\mathbf{w} = \nabla^x \times \mathbf{v}.$$

In view of Definition 2.12 and Result 1.3 we see that $\mathbf{w}$ is the axial vector associated with the skew-symmetric tensor $2\mathbf{W}$. Thus the vorticity at a point is a measure of the rate of rotation or spin at that point. In particular, from the discussion following Stokes' Theorem (Result 2.18), the vorticity at a point in a material body can be interpreted as twice the local angular velocity at that point.

4.5.4 Rigid Motions

A motion φ is called **rigid** if

$$\varphi(\mathbf{X}, t) = \mathbf{c}(t) + \mathbf{R}(t)\mathbf{X}$$

for some time-dependent vector $\mathbf{c}(t)$ and rotation tensor $\mathbf{R}(t)$. For such motions we can show that the spatial velocity field can be written in the form

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t),$$

where $\boldsymbol{\omega}(t)$ is a time-dependent vector called the **spatial angular velocity** of the motion (see Exercise 33). Equivalently, using Result 1.3 we may write

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\Omega}(t)(\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t), \quad (4.18)$$

where $\boldsymbol{\Omega}(t)$ is the skew-symmetric second-order tensor with axial vector $\boldsymbol{\omega}(t)$. From (4.18) we find $\nabla^x \mathbf{v}(\mathbf{x}, t) = \boldsymbol{\Omega}(t)$, from which we deduce $\mathbf{L}(\mathbf{x}, t) = \mathbf{O}$, $\mathbf{W}(\mathbf{x}, t) = \boldsymbol{\Omega}(t)$ and $\mathbf{w}(\mathbf{x}, t) = 2\boldsymbol{\omega}(t)$ for all $t \geq 0$ and $\mathbf{x} \in B_t$. Thus rigid motions produce no rate of strain as measured by $\mathbf{L}$, but produce spin as measured by $\mathbf{W}$ or $\mathbf{w}$.

4.6 Change of Variables

Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body, and for any time $t \geq 0$ consider the deformation φ_t which maps the reference configuration B onto the current configuration B_t . Throughout our developments we will frequently need to transform integrals defined over regions in B_t to integrals over corresponding regions in B . Similarly, we will frequently need to transform integrals defined over surfaces in B_t to integrals defined over corresponding surfaces in B .

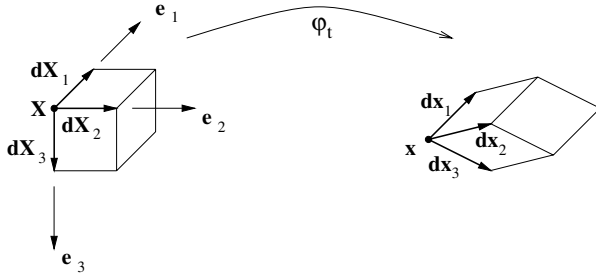


Fig. 4.9 An infinitesimal volume element in B is mapped by the deformation to an infinitesimal volume element in B_t .

4.6.1 Transformation of Volume Integrals

Here we derive a relation between an infinitesimal volume element $dV_{\mathbf{x}}$ at the point $\mathbf{x}$ in B_t , and the corresponding volume element $dV_{\mathbf{X}}$ at the point $\mathbf{X}$ in B (see Figure 4.9). We then use this relation to develop a change of variable formula for volume integrals over B_t and B .

To begin, notice that an infinitesimal volume element $dV_{\mathbf{X}}$ at an arbitrary point $\mathbf{X}$ in B can be represented in terms of a triple scalar product. In particular, let $d\mathbf{X}_1$, $d\mathbf{X}_2$ and $d\mathbf{X}_3$ be three infinitesimal vectors based at the point $\mathbf{X}$. Then the infinitesimal volume element defined by these vectors is

$$dV_{\mathbf{X}} = (d\mathbf{X}_1 \times d\mathbf{X}_2) \cdot d\mathbf{X}_3.$$

As illustrated in Figure 4.9, the three infinitesimal vectors $d\mathbf{X}_k$ at $\mathbf{X}$ are mapped by the deformation φ_t to three infinitesimal vectors $d\mathbf{x}_k$ at $\mathbf{x} = \varphi_t(\mathbf{X})$ in B_t . In particular, we have

$$d\mathbf{x}_k = \nabla^{\mathbf{x}} \varphi(\mathbf{X}, t) d\mathbf{X}_k = \mathbf{F}(\mathbf{X}, t) d\mathbf{X}_k. \quad (4.19)$$

The infinitesimal vectors $d\mathbf{x}_k$ can be used to define a volume element $dV_{\mathbf{x}}$ via the triple scalar product, namely

$$dV_{\mathbf{x}} = (d\mathbf{x}_1 \times d\mathbf{x}_2) \cdot d\mathbf{x}_3. \quad (4.20)$$

Notice that, by properties of the determinant and the triple scalar product, we have

$$(\mathbf{F}\mathbf{u} \times \mathbf{F}\mathbf{v}) \cdot \mathbf{F}\mathbf{w} = (\det \mathbf{F}) \mathbf{u} \times \mathbf{v} \cdot \mathbf{w}, \quad (4.21)$$

for any second-order tensor $\mathbf{F}$ and vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$. Substituting (4.19) into (4.20) and using (4.21) we obtain

$$\begin{aligned} dV_{\mathbf{x}} &= (d\mathbf{x}_1 \times d\mathbf{x}_2) \cdot d\mathbf{x}_3 \\ &= (\mathbf{F}(\mathbf{X}, t)d\mathbf{X}_1 \times \mathbf{F}(\mathbf{X}, t)d\mathbf{X}_2) \cdot \mathbf{F}(\mathbf{X}, t)d\mathbf{X}_3 \\ &= \det \mathbf{F}(\mathbf{X}, t) (d\mathbf{X}_1 \times d\mathbf{X}_2) \cdot d\mathbf{X}_3 \\ &= \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}}. \end{aligned}$$

This expression leads to the following change of variable formula for the transformation of volume integrals.

Result 4.8 Transformation of Volume Integrals. *Let $\phi(\mathbf{x}, t)$ be an arbitrary spatial scalar field on B_t , let Ω_t be an arbitrary subset of B_t , and let Ω be the corresponding subset of B so that $\Omega_t = \boldsymbol{\varphi}_t(\Omega)$. Then*

$$\int_{\Omega_t} \phi(\mathbf{x}, t) dV_{\mathbf{x}} = \int_{\Omega} \phi_m(\mathbf{X}, t) \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}}.$$

□

Remarks:

- (1) The above result provides some insight into the physical meaning of the field $\det \mathbf{F}$. In particular, let $\Omega_{\delta, 0}$ be a ball of radius $\delta > 0$ centered at a point $\mathbf{X}_0 \in B$, and let $\Omega_{\delta, t} = \boldsymbol{\varphi}_t(\Omega_{\delta, 0})$. Then, by Result 4.8 and a Taylor expansion of $\det \mathbf{F}(\mathbf{X}, t)$ about $\mathbf{X}_0$, we have

$$\begin{aligned} \text{vol}(\Omega_{\delta, t}) &= \int_{\Omega_{\delta, t}} dV_{\mathbf{x}} \\ &= \int_{\Omega_{\delta, 0}} \det \mathbf{F}(\mathbf{X}, t) dV_{\mathbf{X}} \\ &= \int_{\Omega_{\delta, 0}} [\det \mathbf{F}(\mathbf{X}_0, t) + \mathcal{O}(\delta)] dV_{\mathbf{X}} \\ &= [\det \mathbf{F}(\mathbf{X}_0, t) + \mathcal{O}(\delta)] \text{vol}(\Omega_{\delta, 0}). \end{aligned}$$

Dividing by $\text{vol}(\Omega_{\delta, 0})$ and taking the limit $\delta \rightarrow 0$ we deduce

$$\det \mathbf{F}(\mathbf{X}_0, t) = \lim_{\delta \rightarrow 0} \frac{\text{vol}(\Omega_{\delta, t})}{\text{vol}(\Omega_{\delta, 0})}.$$

Hence $\det \mathbf{F}(\mathbf{X}_0, t)$ represents the local ratio of deformed volume to reference volume at a point $\mathbf{X}_0 \in B$ under a deformation $\boldsymbol{\varphi}_t$.

- (2) The field defined by $J(\mathbf{X}, t) = \det \mathbf{F}(\mathbf{X}, t)$ is typically called the **Jacobian** field for the deformation φ_t . It is a measure of *volume strain* caused by a deformation φ_t at the point $\mathbf{X}$ at time t . The deformation compresses material in a neighborhood of $\mathbf{X}$ if and only if $J(\mathbf{X}, t) < 1$ (volume is decreased). Similarly, the deformation expands material in a neighborhood of $\mathbf{X}$ if and only if $J(\mathbf{X}, t) > 1$ (volume is increased). When $J(\mathbf{X}, t) = 1$ there is no change in material volume in a neighborhood of $\mathbf{X}$. In this case, the material near $\mathbf{X}$ is distorted in such a way that its volume is preserved. For example, it can be compressed along certain directions, and expanded along others (see also Section 4.7). Notice that $J(\mathbf{X}, t) > 0$ for any admissible deformation.

□

4.6.2 Derivatives of Time-Dependent Integrals

Since the current configuration B_t of a body depends on time, any integral over B_t is generally time-dependent. We will frequently need to compute the time derivatives of such integrals and in this respect the following result will be useful.

Result 4.9 Time Derivative of Jacobian. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and deformation gradient field $\mathbf{F}(\mathbf{X}, t)$. Then*

$$\frac{\partial}{\partial t}(\det \mathbf{F}(\mathbf{X}, t)) = \det \mathbf{F}(\mathbf{X}, t) (\nabla^x \cdot \mathbf{v})(\mathbf{x}, t) \Big|_{\mathbf{x}=\varphi(\mathbf{X}, t)}.$$

□

Proof Using Result 2.23 we find

$$\begin{aligned} \frac{\partial}{\partial t}(\det \mathbf{F}(\mathbf{X}, t)) &= \det \mathbf{F}(\mathbf{X}, t) \mathbf{F}(\mathbf{X}, t)^{-T} : \frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) \\ &= \det \mathbf{F}(\mathbf{X}, t) \operatorname{tr} \left(\mathbf{F}(\mathbf{X}, t)^{-1} \frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) \right) \\ &= \det \mathbf{F}(\mathbf{X}, t) \operatorname{tr} \left(\left\{ \frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) \right\} \mathbf{F}(\mathbf{X}, t)^{-1} \right). \end{aligned} \tag{4.22}$$

By definition of the spatial velocity field $\mathbf{v}$ we have

$$\mathbf{v}(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} = \frac{\partial}{\partial t} \boldsymbol{\varphi}(\mathbf{X}, t) \quad \text{or} \quad v_i(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} = \frac{\partial}{\partial t} \varphi_i(\mathbf{X}, t).$$

We next proceed to express the term $\text{tr}(\{\frac{\partial}{\partial t} \mathbf{F}\} \mathbf{F}^{-1})$ in terms of $\mathbf{v}$. To this end, notice that

$$F_{ij}(\mathbf{X}, t) = \frac{\partial}{\partial X_j} \varphi_i(\mathbf{X}, t)$$

and

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial}{\partial X_j} \varphi_i(\mathbf{X}, t) &= \frac{\partial}{\partial X_j} \frac{\partial}{\partial t} \varphi_i(\mathbf{X}, t) \\ &= \frac{\partial}{\partial X_j} \left(v_i(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \right) \\ &= \frac{\partial}{\partial x_k} v_i(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \frac{\partial}{\partial X_j} \varphi_k(\mathbf{X}, t). \end{aligned}$$

This yields

$$\frac{\partial}{\partial t} F_{ij}(\mathbf{X}, t) = \frac{\partial}{\partial x_k} v_i(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} F_{kj}(\mathbf{X}, t),$$

which in tensor notation becomes

$$\frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) = \nabla^x \mathbf{v}(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \mathbf{F}(\mathbf{X}, t).$$

From this expression we deduce

$$\left\{ \frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) \right\} \mathbf{F}(\mathbf{X}, t)^{-1} = \nabla^x \mathbf{v}(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)},$$

and by taking the trace we obtain

$$\begin{aligned} \text{tr} \left(\left\{ \frac{\partial}{\partial t} \mathbf{F}(\mathbf{X}, t) \right\} \mathbf{F}(\mathbf{X}, t)^{-1} \right) &= \text{tr}(\nabla^x \mathbf{v}(\mathbf{x}, t)) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)} \\ &= (\nabla^x \cdot \mathbf{v})(\mathbf{x}, t) \Big|_{\mathbf{x}=\boldsymbol{\varphi}(\mathbf{X}, t)}. \end{aligned}$$

Substitution of this expression into (4.22) leads to the desired result. $\square$

The above result shows that the time derivative of the Jacobian field depends only on the Jacobian field itself and the divergence of the spatial velocity field. This observation is exploited in the following result, which

provides a convenient formula for the time derivative of volume integrals over arbitrary subsets of B_t .

Result 4.10 Reynolds Transport Theorem. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}(\mathbf{x}, t)$, let Ω_t be an arbitrary volume in B_t with surface denoted by $\partial\Omega_t$ and let $\mathbf{n}$ be the outward unit normal field on $\partial\Omega_t$. Then for any spatial scalar field $\Phi(\mathbf{x}, t)$ we have*

$$\frac{d}{dt} \int_{\Omega_t} \Phi \, dV_{\mathbf{x}} = \int_{\Omega_t} \dot{\Phi} + \Phi(\nabla^x \cdot \mathbf{v}) \, dV_{\mathbf{x}},$$

or equivalently

$$\frac{d}{dt} \int_{\Omega_t} \Phi \, dV_{\mathbf{x}} = \int_{\Omega_t} \frac{\partial}{\partial t} \Phi \, dV_{\mathbf{x}} + \int_{\partial\Omega_t} \Phi \mathbf{v} \cdot \mathbf{n} \, dA_{\mathbf{x}}. \quad (4.23)$$

□

Proof See Exercise 34.

□

Remarks:

- (1) The above result shows that, if the spatial velocity $\mathbf{v}$ is known, then the time derivative of a volume integral over an arbitrary subset Ω_t of B_t can be computed without explicit knowledge of the motion φ or its inverse ψ .
- (2) The term *transport theorem* is motivated by the expression in (4.23). In particular, the rate of change of the integral of Φ over Ω_t is equal to the rate computed as if Ω_t were fixed in its current position, plus the rate at which Φ is *transported* across the boundary $\partial\Omega_t$ by the normal component of velocity $\mathbf{v} \cdot \mathbf{n}$. □

4.6.3 Transformation of Surface Integrals

Let Γ be a surface in B oriented by a unit normal field $\mathbf{N} : \Gamma \rightarrow \mathcal{V}$ and let $\gamma_t = \varphi_t(\Gamma)$ be the corresponding surface in B_t with corresponding orientation given by a unit normal field $\mathbf{n} : \gamma_t \rightarrow \mathcal{V}$. Given a point $\mathbf{X} \in \Gamma$ and an infinitesimal area element $dA_{\mathbf{X}}$ at $\mathbf{X}$, we are interested in how this area element transforms under the mapping φ_t .

For our purposes, we assume that the surface Γ is regular in the sense that, given any point $\mathbf{Z} \in \Gamma$, we can establish a system of **surface**

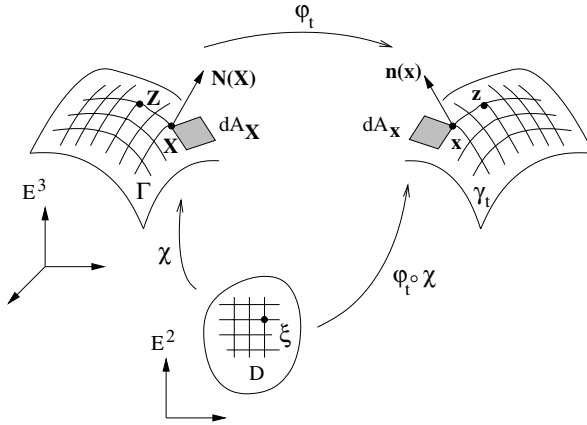


Fig. 4.10 An infinitesimal area element on the surface Γ in B is mapped by the deformation to an infinitesimal area element on the surface γ_t in B_t .

coordinates on Γ near $\mathbf{Z}$. Under this assumption we can find a region $D \subset \mathbb{E}^2$ and a map $\chi : D \rightarrow \mathbb{E}^3$ such that any point $\mathbf{X}$ on Γ near $\mathbf{Z}$ can be written as

$$\mathbf{X} = \chi(\boldsymbol{\xi}), \quad \boldsymbol{\xi} = (\xi_1, \xi_2)^T \in D,$$

as depicted in Figure 4.10. In terms of the surface coordinates $\boldsymbol{\xi}$, an infinitesimal oriented area element in Γ at $\mathbf{X}$ is defined as

$$\mathbf{N}(\mathbf{X}) dA_{\mathbf{X}} = \frac{\partial}{\partial \xi_1} \chi(\boldsymbol{\xi}) \times \frac{\partial}{\partial \xi_2} \chi(\boldsymbol{\xi}) d\xi_1 d\xi_2. \quad (4.24)$$

Under the deformation φ_t , surface coordinates on Γ near $\mathbf{Z}$ become surface coordinates on γ_t near $\mathbf{z} = \varphi_t(\mathbf{Z})$. That is, any point $\mathbf{x}$ on γ_t near $\mathbf{z}$ may be written as

$$\mathbf{x} = \varphi_t(\chi(\boldsymbol{\xi})), \quad \boldsymbol{\xi} = (\xi_1, \xi_2)^T \in D.$$

In terms of the surface coordinates $\boldsymbol{\xi}$, an infinitesimal oriented area element in γ_t at $\mathbf{x}$ is defined as

$$\mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} = \frac{\partial}{\partial \xi_1} \varphi_t(\chi(\boldsymbol{\xi})) \times \frac{\partial}{\partial \xi_2} \varphi_t(\chi(\boldsymbol{\xi})) d\xi_1 d\xi_2. \quad (4.25)$$

Moreover, we have

$$\begin{aligned}\frac{\partial}{\partial \xi_1} \varphi_t(\chi(\xi)) &= \frac{\partial}{\partial \xi_1} \varphi_t^i(\chi(\xi)) e_i \\ &= \left[\frac{\partial}{\partial X_j} \varphi_t^i(X) \right] \Big|_{\mathbf{X}=\chi(\xi)} \frac{\partial}{\partial \xi_1} \chi_j(\xi) e_i \\ &= \mathbf{F}_t(\chi(\xi)) \frac{\partial}{\partial \xi_1} \chi(\xi),\end{aligned}$$

with a similar expression for $\frac{\partial}{\partial \xi_2} \varphi_t(\chi(\xi))$. Using the identity

$$\mathbf{F} \mathbf{u} \times \mathbf{F} \mathbf{v} = (\det \mathbf{F}) \mathbf{F}^{-T} (\mathbf{u} \times \mathbf{v}),$$

which holds for any invertible second-order tensor $\mathbf{F}$ and vectors $\mathbf{u}$ and $\mathbf{v}$ (see Chapter 1, Exercise 17), we can write (4.25) as

$$\begin{aligned}\mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} &= \left[\mathbf{F}_t(\chi(\xi)) \frac{\partial}{\partial \xi_1} \chi(\xi) \right] \times \left[\mathbf{F}_t(\chi(\xi)) \frac{\partial}{\partial \xi_2} \chi(\xi) \right] d\xi_1 d\xi_2 \\ &= [\det \mathbf{F}_t(\chi(\xi))] \mathbf{F}_t(\chi(\xi))^{-T} \left[\frac{\partial}{\partial \xi_1} \chi(\xi) \times \frac{\partial}{\partial \xi_2} \chi(\xi) \right] d\xi_1 d\xi_2.\end{aligned}\tag{4.26}$$

Comparing (4.26) with (4.24) we find

$$\mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} = (\det \mathbf{F}_t(\mathbf{X})) \mathbf{F}_t(\mathbf{X})^{-T} \mathbf{N}(\mathbf{X}) dA_{\mathbf{X}},$$

where $\mathbf{x} = \varphi_t(\mathbf{X})$. This relationship leads to the following result which relates surface integrals between the reference and deformed configurations.

Result 4.11 Transformation of Surface Integrals. *Let $\phi(\mathbf{x}, t)$ be an arbitrary spatial scalar field, $\mathbf{w}(\mathbf{x}, t)$ an arbitrary spatial vector field, and $\mathbf{T}(\mathbf{x}, t)$ an arbitrary spatial second-order tensor field. Consider any region Ω in B which maps to a region Ω_t in B_t under a deformation φ_t . Let $\partial\Omega$ denote the surface of Ω with outward unit normal field $\mathbf{N}(\mathbf{X})$, and let $\partial\Omega_t$ denote the surface of Ω_t with outward unit normal field $\mathbf{n}(\mathbf{x})$. Then*

$$\begin{aligned}\int_{\partial\Omega_t} \phi(\mathbf{x}, t) \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} &= \int_{\partial\Omega} \phi_m(\mathbf{X}, t) \mathbf{G}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) dA_{\mathbf{X}}, \\ \int_{\partial\Omega_t} \mathbf{w}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} &= \int_{\partial\Omega} \mathbf{w}_m(\mathbf{X}, t) \cdot \mathbf{G}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) dA_{\mathbf{X}}, \\ \int_{\partial\Omega_t} \mathbf{T}(\mathbf{x}, t) \mathbf{n}(\mathbf{x}) dA_{\mathbf{x}} &= \int_{\partial\Omega} \mathbf{T}_m(\mathbf{X}, t) \mathbf{G}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) dA_{\mathbf{X}},\end{aligned}$$

where $\mathbf{G}$ is a material second-order tensor field defined as

$$\mathbf{G}(\mathbf{X}, t) = (\det \mathbf{F}(\mathbf{X}, t)) \mathbf{F}(\mathbf{X}, t)^{-T}.$$

□

Notice that other types of surface integrals can also be considered and transformed in a similar way (see Exercise 35).

4.7 Volume-Preserving Motions

A motion $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ of a continuum body is called **volume-preserving** or **isochoric** if $\text{vol}[\varphi_t(\Omega)] = \text{vol}[\Omega]$ for all $t \geq 0$ and all subsets $\Omega \subseteq B$. In particular, a motion is volume-preserving if the volume of the body, and each of its parts, remains constant throughout a motion.

Result 4.12 Conditions for Volume Preservation. *Let $\varphi : B \times [0, \infty) \rightarrow \mathbf{E}^3$ be a motion of a continuum body with spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and deformation gradient field $\mathbf{F}(\mathbf{X}, t)$. Then φ is volume-preserving if and only if $\det \mathbf{F}(\mathbf{X}, t) = 1$ for all $\mathbf{X} \in B$ and $t \geq 0$, or equivalently $(\nabla^x \cdot \mathbf{v})(\mathbf{x}, t) = 0$ for all $\mathbf{x} \in B_t$ and $t \geq 0$.*

□

Proof See Exercise 36.

□

Simple motions such as translations and rotations of body are volume-preserving because they produce no distortion. However, motions which do distort a body can also be volume-preserving (see Exercise 37).

Bibliographic Notes

Much of the material presented here on deformations, motions, strain and rate of strain can be found in the text by Gurtin (1981). A detailed analysis of these concepts in general coordinate systems is given in Ogden (1984), and a similar detailed account from a more engineering point of view can be found in Malvern (1969). The treatise by Truesdell and Toupin (1960) contains a wealth of information on these concepts and their history. For a mathematical analysis of deformations and motions within the context of Riemannian geometry and manifold theory see Marsden and Hughes (1983).

Constrained motions such as rigid and volume-preserving motions are of special interest in continuum mechanics. For example, rigid motions

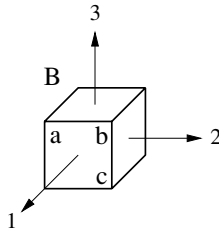
are of interest in the study of rigid bodies, and provide a standard against which other motions can be compared. Volume-preserving motions are of interest in the study of incompressible fluids (see Chapter 6) and solids. A complete mathematical characterization of rigid deformations and motions can be found in Gurtin (1981). For a more rigorous treatment of rigid deformations see Ciarlet (1988, Volume 1), and for a discussion of constrained motions other than rigid and volume-preserving see Ogden (1984) and Antman (1995).

The change of variable formulae for volume and surface integrals play a fundamental role in continuum mechanics and will be used repeatedly in later chapters. Discussions of these formulae can be found in standard texts on advanced multivariable calculus, for example Bourne and Kendall (1992) and Marsden and Tromba (1988). Rigorous proofs of the change of variable formulae can be found in the text by Munkres (1991).

The Reynolds Transport Theorem is of special importance in the study of fluids. In particular, it forms the basis for the so-called *control volume* method for studying fluid flow. For applications of the transport theorem in this context see White (2006) and Gurtin (1981).

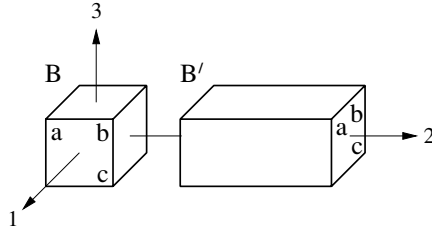
Exercises

- 4.1 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid |X_i| < 1\}$ be the reference configuration of a body as shown below, and consider the deformation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X})$ defined in components by $x_1 = X_1$, $x_2 = X_3 + 4$, $x_3 = -X_2$.



- (a) Sketch the deformed configuration $B' = \boldsymbol{\varphi}(B)$ relative to the same coordinate frame and indicate the points in B' which correspond to the vertices a , b , c in B .
- (b) Find the components of the displacement field $\mathbf{u}$.
- 4.2 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid |X_i| < 1\}$ and $B' = \{\mathbf{x} \in \mathbb{E}^3 \mid |x_1| < 1, |x_3| < 1, |x_2 - 6| < 3\}$ be the reference and deformed configurations of a body as shown below, and consider a deformation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X})$ of the form $x_1 = pX_2 + q$, $x_2 = rX_1 + s$, $x_3 = X_3$.

By comparing the faces of B' and B find constants p, q, r, s such that $B' = \varphi(B)$.



- 4.3 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid |X_i| < 1\}$ be the reference configuration of a body. For each deformation $\mathbf{x} = \varphi(\mathbf{X})$ given below (i) sketch the deformed configuration B' , (ii) find the components of $\nabla\varphi$, (iii) determine if φ is one-to-one, and (iv) determine if φ is admissible:

(a) $x_1 = 1 + X_1, \quad x_2 = \frac{1}{2}X_2 + 5, \quad x_3 = 2X_3,$

(b) $x_1 = X_1, \quad x_2 = 3, \quad x_3 = X_3,$

(c) $x_1 = X_1, \quad x_2 = X_2, \quad x_3 = 4 - X_3.$

- 4.4 Consider a deformation $\mathbf{x} = \varphi(\mathbf{X})$ defined in components by

$$x_1 = X_1^2, \quad x_2 = X_3^2, \quad x_3 = X_2X_3.$$

(a) Find the components of $\nabla\varphi$ and determine $\det \nabla\varphi$.

(b) Is φ admissible for arbitrary B ?

- 4.5 For each deformation $\mathbf{x} = \varphi(\mathbf{X})$ given below find the components of the deformation gradient $\mathbf{F}$ and determine if φ is homogeneous or non-homogeneous:

(a) $x_1 = X_1, \quad x_2 = X_2X_3, \quad x_3 = X_3 - 1,$

(b) $x_1 = 2X_2 - 1, \quad x_2 = X_3, \quad x_3 = 3 + 5X_1,$

(c) $x_1 = \exp(X_1), \quad x_2 = -X_3, \quad x_3 = X_2.$

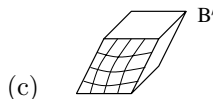
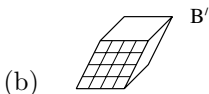
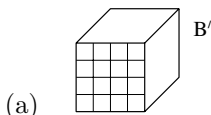
- 4.6 Let φ be a homogeneous deformation with deformation gradient $\mathbf{F}$. For any two points $\mathbf{X}$ and $\mathbf{Y}$ show that

$$\varphi(\mathbf{X}) = \varphi(\mathbf{Y}) + \mathbf{F}(\mathbf{X} - \mathbf{Y}).$$

- 4.7 Let $\varphi : B \rightarrow B'$ be a homogeneous deformation with deformation gradient $\mathbf{F}$, and let $\mathbf{X}(\sigma) = \mathbf{X}_0 + \sigma \mathbf{v}$ be a line segment through the point $\mathbf{X}_0$ in B with direction $\mathbf{v}$. Show that $\varphi(\mathbf{X}(\sigma))$ is a line segment through the point $\varphi(\mathbf{X}_0)$ in B' with direction $\mathbf{F}\mathbf{v}$.
- 4.8 One way to visualize a deformation is to show how curves etched or marked on the surface of a body appear before and after the deformation. Suppose a reference configuration B is marked with lines as follows:



Which of the deformations visualized below appear to be homogeneous? Which appear to be non-homogeneous?



- 4.9 Prove Result 4.1.
- 4.10 Prove Result 4.2.
- 4.11 Prove Result 4.3.
- 4.12 Let $\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$ be the right and left polar decompositions of a deformation gradient $\mathbf{F}$. Show that:
- (a) $\mathbf{U}$ and $\mathbf{V}$ have the same eigenvalues,
 - (b) if $\{\mathbf{u}_i\}$ is a frame of eigenvectors of $\mathbf{U}$, then $\{\mathbf{R}\mathbf{u}_i\}$ is a frame of eigenvectors of $\mathbf{V}$. Thus, in general, $\mathbf{U}$ and $\mathbf{V}$ have different eigenvectors.
- 4.13 Consider a deformation φ defined in components by

$$[\varphi(\mathbf{X})] = \begin{Bmatrix} pX_1 + a \\ qX_2 + b \\ rX_3 + c \end{Bmatrix},$$

where p, q, r, a, b, c are constants.

- (a) Find conditions on p, q, r for φ to be an admissible deformation of $B = \mathbf{E}^3$.
- (b) Given an arbitrary point $\mathbf{Y}$ find the components of $\mathbf{d}, \mathbf{s}$ and

$\mathbf{r}$ such that $\boldsymbol{\varphi} = \mathbf{r} \circ \mathbf{s} \circ \mathbf{d}$, where $\mathbf{d}$ is a translation, $\mathbf{s}$ is a stretch from $\mathbf{Y}$ and $\mathbf{r}$ is rotation about $\mathbf{Y}$.

- 4.14 Let $\mathbf{Y}$ be an arbitrary point and let $\mathbf{Q}$ be an arbitrary rotation tensor and consider the deformation

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{Y} + \mathbf{Q}(\mathbf{X} - \mathbf{Y}).$$

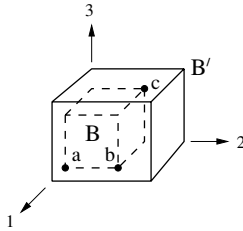
In particular, $\boldsymbol{\varphi}$ is a rotation about $\mathbf{Y}$. Find the deformation gradient $\mathbf{F}$ and the Cauchy–Green strain tensor $\mathbf{C}$. Does $\mathbf{F}$ depend on $\mathbf{Q}$? What about $\mathbf{C}$?

- 4.15 Suppose the deformation gradient at a point $\mathbf{X}_0$ in a body has components

$$[\mathbf{F}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Find the components of the Cauchy–Green strain tensor $\mathbf{C}$ and the right stretch tensor $\mathbf{U}$.

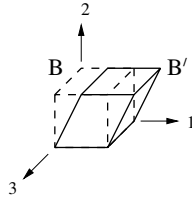
- 4.16 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid 0 < X_i < 1\}$ and consider a deformation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X})$ of the form $x_1 = pX_1$, $x_2 = qX_2$, $x_3 = rX_3$, where p, q, r are constants. Notice that, because $\boldsymbol{\varphi}$ is homogeneous, the Cauchy–Green strain tensor $\mathbf{C}$ will be constant.



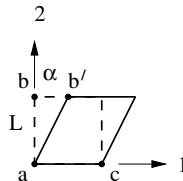
- (a) Find the components of $\mathbf{C}$.

(b) Find the stretch λ in (i) the direction parallel to the edge $\overline{ab}$ and (ii) the direction parallel to the diagonal $\overline{ac}$.

- 4.17 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid 0 < X_i < 1\}$ and consider a deformation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X})$ of the form $x_1 = X_1 + \alpha X_2$, $x_2 = X_2$, $x_3 = X_3$, where $\alpha > 0$ is a constant. Such a deformation is called a **simple shear** in the $\mathbf{e}_1, \mathbf{e}_2$ -plane.



- (a) Find the components of the Cauchy–Green strain tensor $\mathbf{C}$.
- (b) Find the shear γ for the direction pair $(\mathbf{e}_1, \mathbf{e}_2)$ and for the pair $(\mathbf{e}_1, \mathbf{e}_3)$. What happens to these shears in the limits $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$?
- (c) Find the extreme values of the stretch λ and the directions in which these occur. Do the directions of extreme stretch correspond to any of the diagonal directions $\pm \frac{1}{\sqrt{2}}(\mathbf{e}_1 + \mathbf{e}_2)$ or $\pm \frac{1}{\sqrt{2}}(\mathbf{e}_1 - \mathbf{e}_2)$ in the $\mathbf{e}_1, \mathbf{e}_2$ -plane?
- 4.18 Show that the shear $\gamma(\mathbf{e}, \mathbf{d})$ between any two right principal directions $\mathbf{e}$ and $\mathbf{d}$ is zero.
- 4.19 Prove Result 4.5.
- 4.20 Let $\mathbf{c}$ be an arbitrary vector, $\mathbf{A}$ an arbitrary tensor and ϵ an arbitrary scalar. Supposing the components of $\mathbf{c}$ and $\mathbf{A}$ are of order unity and $0 \leq \epsilon \ll 1$, determine which of the following deformations are small:
- (a) $\varphi(\mathbf{X}) = \mathbf{X} + \mathbf{c}$,
- (b) $\varphi(\mathbf{X}) = \mathbf{A}\mathbf{X} + \mathbf{c}$,
- (c) $\varphi(\mathbf{X}) = \epsilon\mathbf{A}\mathbf{X} + \mathbf{c}$,
- (d) $\varphi(\mathbf{X}) = (\mathbf{I} + \epsilon\mathbf{A})\mathbf{X} + \mathbf{c}$.
- 4.21 Let $B = \{\mathbf{X} \in \mathbf{E}^3 \mid 0 < X_i < L\}$ and consider the simple shear deformation $\mathbf{x} = \varphi(\mathbf{X})$ given by $x_1 = X_1 + (\alpha/L)X_2$, $x_2 = X_2$, $x_3 = X_3$, where $\alpha > 0$ is a constant.



(a) Show that $\boldsymbol{\varphi}$ is small when $\alpha \ll L$ and find the components of the infinitesimal strain tensor $\mathbf{E}$.

(b) Use the series expansion $\sqrt{1 + \epsilon^2} = 1 + \frac{\epsilon^2}{2} - \frac{\epsilon^4}{8} + \dots$ to show that

$$\frac{\overline{ab'} - \overline{ab}}{\overline{ab}} = \frac{(\alpha/L)^2}{2} - \frac{(\alpha/L)^4}{8} + \dots.$$

(Here $\overline{ab}$ denotes length.) Is this result consistent with the interpretation of E_{22} and its value found in part (a)?

(c) Use the series expansion $\arctan(\epsilon) = \epsilon - \frac{\epsilon^3}{3} + \frac{\epsilon^5}{5} - \dots$ to show that

$$\angle bac - \angle b'ac = (\alpha/L) - \frac{(\alpha/L)^3}{3} + \frac{(\alpha/L)^5}{5} - \dots.$$

(Here $\angle bac$ denotes angle.) Is this result consistent with the interpretation of E_{12} and its value found in part (a)?

4.22 Consider a rigid deformation of the form

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{R}\mathbf{X} + \mathbf{c},$$

where $\mathbf{R}$ is a small rotation in the sense that $\mathbf{R} = \mathbf{I} + \mathcal{O}(\epsilon)$ for some $0 \leq \epsilon \ll 1$. In particular, suppose

$$\mathbf{R} = \exp(\epsilon \mathbf{W}) = \mathbf{I} + \epsilon \mathbf{W} + \frac{\epsilon^2}{2} \mathbf{W}^2 + \dots,$$

where $\mathbf{W}$ is an arbitrary skew-symmetric tensor with components of order unity (see Result 1.7).

(a) Find the displacement field $\mathbf{u}$ and show that $\boldsymbol{\varphi}$ is a small deformation for any reference configuration B .

(b) Show that, if terms of order $\mathcal{O}(\epsilon^2)$ are neglected, then $\boldsymbol{\varphi}$ has the form of an infinitesimally rigid deformation.

4.23 Let $B = \{\mathbf{X} \in \mathbb{E}^3 \mid 0 < X_i < 1\}$ and consider the motion $\boldsymbol{\varphi}$ defined in components by

$$[\boldsymbol{\varphi}(\mathbf{X}, t)] = \begin{Bmatrix} \exp(tX_1) + X_1 - 1 \\ \exp(tX_2) + X_2 - 1 \\ \exp(tX_3) + X_3 - 1 \end{Bmatrix}.$$

Determine the position of the boundary points $[\mathbf{X}_0] = (0, 0, 0)^T$ and $[\mathbf{X}_1] = (1, 1, 1)^T$ as a function of t .

- 4.24 Let $B = \mathbf{E}^3$ and consider the motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ defined by

$$x_1 = e^t X_1 + X_3, \quad x_2 = X_2, \quad x_3 = X_3 - tX_1.$$

- (a) Show that the inverse motion $\mathbf{X} = \boldsymbol{\psi}(\mathbf{x}, t)$ is given by

$$X_1 = \frac{x_1 - x_3}{t + e^t}, \quad X_2 = x_2, \quad X_3 = \frac{tx_1 + e^t x_3}{t + e^t}.$$

- (b) Verify that $\boldsymbol{\varphi}(\boldsymbol{\psi}(\mathbf{x}, t), t) = \mathbf{x}$ and $\boldsymbol{\psi}(\boldsymbol{\varphi}(\mathbf{X}, t), t) = \mathbf{X}$.

- 4.25 Consider the motion in Exercise 24 and let $\Omega(\mathbf{X}, t)$ and $\Gamma(\mathbf{x}, t)$ be material and spatial fields defined by

$$\Omega(\mathbf{X}, t) = X_1 + t, \quad \Gamma(\mathbf{x}, t) = x_1 + t.$$

- (a) Find the spatial description Ω_s of Ω .

- (b) Find the material description Γ_m of Γ .

- 4.26 Consider the motion from Exercise 24 and consider the spatial field $\Gamma(\mathbf{x}, t) = x_1 + t$.

- (a) Find the material time derivative $\dot{\Gamma}(\mathbf{x}, t)$.

- (b) Find the partial time derivative $\frac{\partial}{\partial t}\Gamma(\mathbf{x}, t)$ and verify that $\dot{\Gamma}(\mathbf{x}, t) \neq \frac{\partial}{\partial t}\Gamma(\mathbf{x}, t)$.

- 4.27 Let $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ be the motion considered in Exercise 24.

- (a) Find the components of the material velocity $\mathbf{V}(\mathbf{X}, t)$ and the spatial velocity $\mathbf{v}(\mathbf{x}, t)$.

- (b) Find the components of the material acceleration $\mathbf{A}(\mathbf{X}, t)$ and the spatial acceleration $\mathbf{a}(\mathbf{x}, t)$.

- (c) Verify that $\mathbf{a}(\mathbf{x}, t) \neq \frac{\partial}{\partial t}\mathbf{v}(\mathbf{x}, t)$.

- 4.28 Let $B = \{\mathbf{X} \in \mathbf{E}^3 \mid 0 < X_i < 1\}$ and consider the motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ defined by

$$x_1 = tX_2^2 + X_1, \quad x_2 = tX_1 + X_2, \quad x_3 = tX_3 + X_3.$$

- (a) Find the components of the deformation gradient and find the largest number $t_c > 0$ for which $\det \mathbf{F}(\mathbf{X}, t) > 0$ for all $\mathbf{X} \in B$ and $t \in [0, t_c]$.

- (b) Find the components of the inverse motion $\mathbf{X} = \boldsymbol{\psi}(\mathbf{x}, t)$ for $t \in [0, t_c]$.

(c) Find the components of the material velocity field $\mathbf{V}(\mathbf{X}, t)$ and the spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ for $t \in [0, t_c]$.

4.29 Let $B = \mathbb{E}^3$ and consider the motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ defined by

$$x_1 = (1 + t)X_1, \quad x_2 = X_2 + tX_3, \quad x_3 = X_3 - tX_2.$$

Moreover, consider the spatial field $\phi(\mathbf{x}, t) = tx_1 + x_2$.

(a) Show that $\det \mathbf{F}(\mathbf{X}, t) > 0$ for all $t \geq 0$ and find the components of the inverse motion $\mathbf{X} = \boldsymbol{\psi}(\mathbf{x}, t)$ for all $t \geq 0$.

(b) Find the components of the spatial velocity field $\mathbf{v}(\mathbf{x}, t)$.

(c) Find the material time derivative of ϕ using the definition $\dot{\phi} = [\dot{\phi}_m]_s$.

(d) Find the material time derivative of ϕ using Result 4.7. Do you obtain the same result as in part (c)?

4.30 Consider the deformation $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ given by

$$\begin{aligned} x_1 &= \cos(\omega t)X_1 + \sin(\omega t)X_2, \\ x_2 &= -\sin(\omega t)X_1 + \cos(\omega t)X_2, \\ x_3 &= (1 + \alpha t)X_3. \end{aligned}$$

Notice that this deformation corresponds to rotation (with rate ω) in the $\mathbf{e}_1, \mathbf{e}_2$ -plane together with extension (with rate α) along the $\mathbf{e}_3$ -axis.

(a) Find the components of the inverse motion $\mathbf{X} = \boldsymbol{\psi}(\mathbf{x}, t)$.

(b) Find the components of the spatial velocity field $\mathbf{v}(\mathbf{x}, t)$.

(c) Find the components of the rate of strain and spin tensors $\mathbf{L}(\mathbf{x}, t)$ and $\mathbf{W}(\mathbf{x}, t)$. Verify that $\mathbf{L}$ is determined by α , whereas $\mathbf{W}$ is determined by ω .

4.31 Consider a motion $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbb{E}^3$. For any fixed $t \geq 0$ let $\mathbf{v}$ be the spatial velocity field in the current configuration B_t and let $\boldsymbol{\psi}$ be the inverse motion. For any $s > 0$ let $\hat{\boldsymbol{\varphi}}_s : B_t \rightarrow B_{t+s}$ be the motion which coincides with $\boldsymbol{\varphi}_{t+s} : B \rightarrow B_{t+s}$ in the sense that

$$\hat{\boldsymbol{\varphi}}(\mathbf{x}, s) = \boldsymbol{\varphi}(\mathbf{X}, t + s)|_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)}, \quad \forall \mathbf{x} \in B_t.$$

(a) Show that $\hat{\boldsymbol{\varphi}}(\mathbf{x}, 0) = \mathbf{x}$ for all $\mathbf{x} \in B_t$.

(b) Show that $\frac{\partial}{\partial s}\hat{\boldsymbol{\varphi}}(\mathbf{x}, 0) = \mathbf{v}(\mathbf{x}, t)$ for all $\mathbf{x} \in B_t$.

(c) Let $\hat{\mathbf{F}}(\mathbf{x}, s) = \nabla^x \hat{\boldsymbol{\varphi}}(\mathbf{x}, s)$ be the deformation gradient associated with $\hat{\boldsymbol{\varphi}}$. Show that $\hat{\mathbf{F}}(\mathbf{x}, 0) = \mathbf{I}$ and $\frac{\partial}{\partial s}\hat{\mathbf{F}}(\mathbf{x}, 0) = \nabla^x \mathbf{v}(\mathbf{x}, t)$.

(d) Let $\hat{\mathbf{E}} = \text{sym}(\nabla^x \hat{\mathbf{u}}) = \text{sym}(\hat{\mathbf{F}} - \mathbf{I})$ be the infinitesimal strain tensor associated with $\hat{\boldsymbol{\varphi}}$. Show that

$$\mathbf{L}(\mathbf{x}, t) = \frac{\partial}{\partial s}\hat{\mathbf{E}}(\mathbf{x}, 0).$$

(e) Consider the right polar decomposition $\hat{\mathbf{F}} = \hat{\mathbf{R}}\hat{\mathbf{U}}$, where $\hat{\mathbf{U}}^2 = \hat{\mathbf{F}}^T \hat{\mathbf{F}}$. Show that

$$\mathbf{L}(\mathbf{x}, t) = \frac{\partial}{\partial s}\hat{\mathbf{U}}(\mathbf{x}, 0), \quad \mathbf{W}(\mathbf{x}, t) = \frac{\partial}{\partial s}\hat{\mathbf{R}}(\mathbf{x}, 0).$$

4.32 Consider a motion $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ with spatial velocity field $\mathbf{v}$, spatial acceleration field $\mathbf{a}$, spatial vorticity field $\mathbf{w}$ and spatial spin field $\mathbf{W}$.

(a) Show that $2\mathbf{W}\mathbf{c} = \mathbf{w} \times \mathbf{c}$ for any arbitrary vector $\mathbf{c}$. (Thus $\mathbf{w}$ is the axial vector of $2\mathbf{W}$.)

(b) Show that the acceleration field satisfies

$$\mathbf{a} = \frac{\partial \mathbf{v}}{\partial t} + \mathbf{w} \times \mathbf{v} + \nabla^x \left(\frac{1}{2} |\mathbf{v}|^2 \right).$$

4.33 Consider an arbitrary rigid motion $\boldsymbol{\varphi} : B \times [0, \infty) \rightarrow \mathbf{E}^3$ of the form

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{R}(t)\mathbf{X} + \mathbf{c}(t),$$

where $\mathbf{R}(t)$ is a rotation tensor and $\mathbf{c}(t)$ is a vector.

(a) Find the inverse motion $\boldsymbol{\psi}(\mathbf{x}, t)$.

(b) Let $\boldsymbol{\Omega}(t) = \dot{\mathbf{R}}(t)\mathbf{R}(t)^T$. Show that $\boldsymbol{\Omega}(t)$ is skew-symmetric.

(c) Show that the spatial velocity field can be written in the form

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\Omega}(t)(\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t).$$

4.34 Prove Result 4.10.

4.35 Show that

$$\int_{\partial\Omega_t} \mathbf{w}(\mathbf{x}, t) \otimes \mathbf{n}(\mathbf{x}) \, dA_{\mathbf{x}} = \int_{\partial\Omega} \mathbf{w}_m(\mathbf{X}, t) \otimes \mathbf{G}(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) \, dA_{\mathbf{X}},$$

where the tensor $\mathbf{G}$ and unit normals $\mathbf{n}$ and $\mathbf{N}$ are as defined in Result 4.11.

4.36 Prove Result 4.12.

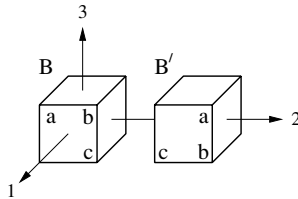
4.37 Let $B = \{\mathbf{X} \in \mathbf{E}^3 \mid 0 < X_i < 1\}$ and consider the simple shear motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ defined by

$$x_1 = X_1 + \alpha t X_2, \quad x_2 = X_2, \quad x_3 = X_3,$$

where $\alpha > 0$ is a constant. Show that $\boldsymbol{\varphi}(\mathbf{X}, t)$ is volume-preserving.

Answers to Selected Exercises

4.1 (a) Consider an arbitrary line segment in B in the direction of $\mathbf{e}_3$, namely $\mathbf{X}(\sigma) = \alpha \mathbf{e}_1 + \beta \mathbf{e}_2 + \sigma \mathbf{e}_3$, where α and β are fixed and $\sigma \in (-1, 1)$ is the coordinate along the segment. Under deformation, this segment becomes $\mathbf{x}(\sigma) = \boldsymbol{\varphi}(\mathbf{X}(\sigma)) = \alpha \mathbf{e}_1 + (\sigma + 4) \mathbf{e}_2 - \beta \mathbf{e}_3$, which corresponds to a line segment in B' in the direction of $\mathbf{e}_2$. Notice that the line segment in B is centered at the point with coordinates $(\alpha, \beta, 0)$, and the corresponding line segment in B' , which is of the same length, is centered at the point with coordinates $(\alpha, 4, -\beta)$. By considering different values for α and β , for example values corresponding to the vertical edges of B , we obtain the following sketch.



(b) $u_1 = 0$, $u_2 = X_3 + 4 - X_2$, $u_3 = -X_2 - X_3$.

4.3 In general, a deformation $\boldsymbol{\varphi} : B \rightarrow B'$ is one-to-one if and only if the equation $\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{x}$ has a unique solution $\mathbf{X}$ for each $\mathbf{x}$ in the range of $\boldsymbol{\varphi}$, which is equal to B' since $\boldsymbol{\varphi}$ is onto. For homogeneous deformations of the form $\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{F}\mathbf{X} + \mathbf{v}$, the